

BCA-205 OPERATING SYSTEM & UNIX

UNIT 1: INTRODUCTION TO OPERATING SYSTEMS – CHEAT SHEET

READY FOR
9 JULY TEST

1. INTRODUCTION TO OPERATING SYSTEMS

Definition (English):

An Operating System (OS) is system software that acts as an interface between the user and hardware. It manages all computer resources like CPU, memory, I/O devices, files, programs and provides services for users and programs.

Hinglish (Simple):

OS ek manager ki tarah hota hai jo hardware aur user ke beech ka kaam karta hai. Ye decide karta hai ki kaun program kab chalega, kisko kitna CPU time milega, kaun sa device use hoga, data kaha store hoga etc.

Real Life Example:

Jaise ek School Principal sab teachers, students, class rooms, timetable, resources ko manage karta hai. Waise hi OS sab computer resources ko manage karta hai.

PYQ Indicator:

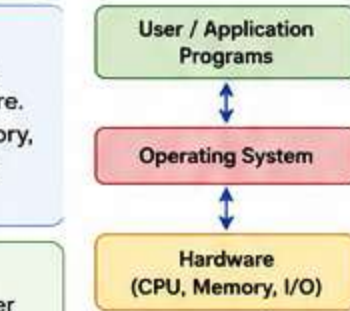
- 2014 (2M): What is the role of operating system?
- 2016 (2M): What resources must be protected by OS?
- 2018 (12M): Discuss services provided by OS.
- 2020 (10M): Explain OS with diagram.

2-Mark Ready Answer:

Operating System is system software that acts as an interface between user and hardware. It manages resources like CPU, memory, I/O devices and files. It provides services for program execution, I/O operations and protection.

Long Answer Key Points (12-Mark):

- Definition and need of OS
- Functions / Services of OS
- Types of OS (Batch, Multiprogramming, Time-sharing, Real-time, Multiprocessor)
- OS structure (User – OS – Hardware)
- Conclusion



Short Trick Box:

OS = Oversee (manage) System
User se request aaya → OS ne resource diya → Kaam ho gaya.

One Line Summary:

OS ek manager hai jo user, programs aur hardware ke beech resources ko efficiently manage karta hai.

4. TIME-SHARING SYSTEM

Definition (English):

In time-sharing system, CPU time is divided into small time slices (time quantum) and shared among multiple users iteratively.

Hinglish:

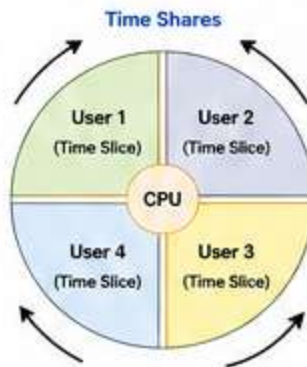
Yahaan har user ko thoda-thoda time (time slice) milta hai aur system aisa lagta hai jaise sabko ek saath service mil rahi ho.

Example:

Jaise IRCTC website par hazaron log alag-alag time me thoda-thoda server time paate hain.

Features:

- Fast response time
- Interactive system
- Used in multi-user systems



6. SYSTEM CALLS

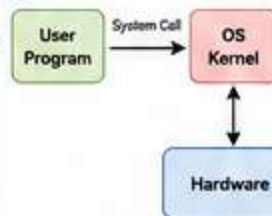
System call is the way by which a user program requests a service from OS kernel.

Hinglish:

User program jab OS se koi service (jaise file open, read, write, process create) mangta hai, to system call ke through request bhejta hai.

Types of System Calls:

- Process Control: create(), exit(), fork()
- File Management: open(), close(), read(), write()
- Device Management: read(), write()
- Information Maintenance: time(), date(), getpid()
- Communication: pipe(), shmget(), socket()



8. BASIC CONCEPTS OF CPU SCHEDULING

Definition:

Scheduling is the task of selecting a process from the ready queue and allocating CPU to it.

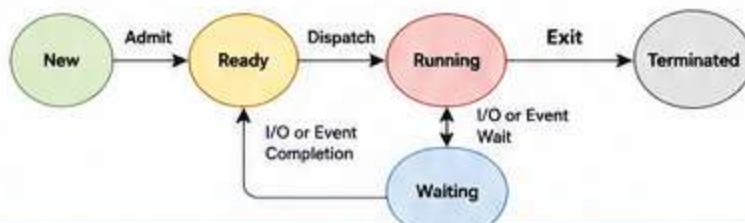
Scheduling Queue:

- Job Queue: All processes in the system
- Ready Queue: Processes ready to execute

CPU Scheduling Levels:

- Long-term (Job Scheduler)
- Medium-term (Swapper)
- Short-term (CPU Scheduler)

Process State Diagram



11. ALGORITHM EVALUATION

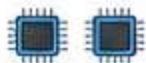
Evaluate algorithms on basis of:

1. CPU Utilization
2. Throughput
3. Turnaround Time
4. Waiting Time
5. Response Time

12. MULTIPLE PROCESSOR SCHEDULING

Two types:

1. Symmetric Multiprocessing (SMP): All CPUs are identical and share memory.
2. Asymmetric Multiprocessing (AMP): Master-Slave relationship.



13. REAL TIME SCHEDULING

Used in time-critical systems where deadlines are important.

Types:

- Hard Real-time
- Soft Real-time

2. OPERATING SYSTEM SERVICES

OS users aur programs ko kuch important services provide karta hai.

Service	Use
Program Execution	Program ko memory me load karke execute karta hai.
I/O Operations	Input (keyboard) ya Output (printer) operations ko handle karta hai.
File System Manipulation	Files create, delete, read, write, copy, rename karta hai.
Communication	Processes ke beech data exchange allow karta hai.
Error Detection	Hardware ya program errors ko detect karta hai.
Resource Allocation	CPU time, memory, I/O devices allocate karta hai.
Protection & Security	Unauthorized access se resources ko protect karta hai.

Short Trick:

P I F C E R P → Program, I/O, File, Communication, Error, Resource, Protection

3. MULTIPROGRAMMING

Definition (English):

Multiprogramming is a technique in which multiple programs are kept in memory at the same time and CPU is shared among them.

Hinglish:

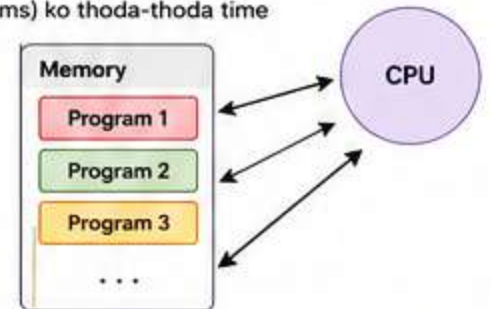
Jab ek time par multiple programs memory me rehte hain aur CPU switch karke sabko time deta hai, use multiprogramming kehte hain.

Example:

Jaise ek chef (CPU) alag-alag orders (programs) ko thoda-thoda time de kar banata hai.

Benefits:

- CPU Utilization badhta hai
- Throughput badhta hai
- CPU idle time kam hota hai



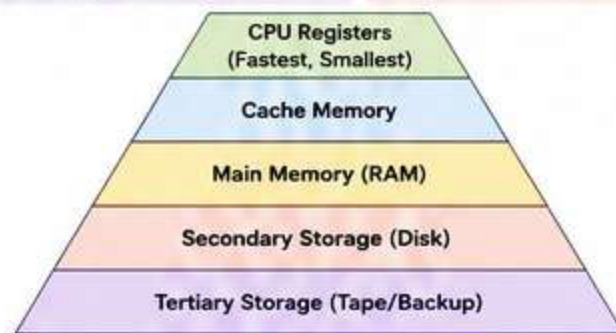
5. STORAGE STRUCTURES

Primary Storage (Main Memory)

- Directly accessible by CPU
- Fast but small in size
- Types: RAM, ROM

Secondary Storage (Auxiliary)

- Not directly accessible by CPU
- Large but slower
- Types: HDD, SSD, CD, Pen Drive



Top = Fast
But Costly
& Small

Bottom =
Slow But
Cheaper &
Large

7. MULTIPROCESSOR SYSTEM

Definition (English):

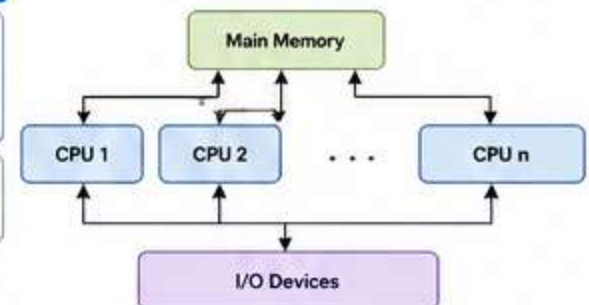
A system with two or more CPUs (shared memory) working together is called multiprocessor system.

Hinglish:

Jab system me 2 ya zyada CPU hote hain jo ek saath kaam karte hain.

Advantages:

- High Reliability
- More Throughput
- Economy of Scale



9. SCHEDULING CRITERIA

Criteria	Goal
CPU Utilization	Maximize
Throughput	Maximize
Turnaround Time	Minimize
Waiting Time	Minimize
Response Time	Minimize

10. SCHEDULING ALGORITHMS (Types)

- FCFS (First Come First Serve)
- SJF (Shortest Job First)
- Priority Scheduling
- Round Robin Scheduling
- Multilevel Queue Scheduling
- Multilevel Feedback Queue
- Real Time Scheduling (Rate Monotonic, Earliest Deadline First)

14. I/O DEVICES ORGANIZATION & BUFFERING

I/O Organization: OS controls I/O devices through device drivers.

I/O Buffering:

- Single Buffering
- Circular Buffering
- Double Buffering
- Spooling

(Use buffer to reduce CPU wait time)

PYQ HINTS (Unit 1)

Important Questions Repeated Many Times: Role of OS, OS Services, Multiprogramming, Time Sharing, System Calls, Scheduling Algorithms, Scheduling Criteria, Paging (just concept), Real Time Scheduling.

★ FOCUS MORE ON
DEFINITIONS + DIAGRAMS

BCA-205 OPERATING SYSTEM & UNIX

UNIT 2: PROCESS MANAGEMENT – CHEAT SHEET

READY FOR
9 JULY TEST

1. PROCESS CONCEPT

Definition (English):

A process is a program in execution. It includes the program code, current activity, program counter, registers, variables and stack.

Hinglish (Simple):

Process matlab CPU par chal raha program. Jab aap koi app open karte ho, wo memory me load hota hai aur CPU usko run karta hai, tab wo process kehata hai.

Real Life Example:

Jab aap WhatsApp open karte ho, to uska program memory me load hota hai aur CPU usko chalata hai. Ye hi ek process hai.

PYQ Indicator:

- 2014 (2M): Describe about process.
- 2016 (2M): What is meant by 'automatic job sequencing'?
- 2018 (12M): Explain process and its states with diagram.
- 2020 (10M): Discuss operations on process.

2-Mark Ready Answer:

A process is a program in execution. It has its own program code, data, stack, heap, CPU registers and program counter. OS manages all processes.

Long Answer Key Points (12-Mark):

- Definition of Process
- Components of a Process
- Process State Diagram
- Process Control Block (PCB)
- Operations on Process
- Process Creation
- Process Termination
- Benefits of Process Concept

Short Trick Box

P C D S H R = Process ke parts
Program, Counter, Data, Stack,
Heap, Registers

Process in Memory

Program Code

Data Section

Heap

Stack

Program Counter

CPU Registers

2. PROCESS SCHEDULING

Definition (English):

Process scheduling is the activity of selecting a process from the ready queue and allocating CPU to it.

Hinglish (Simple):

Scheduler decide karta hai ki ready queue me jo processes hain, unme se kaun sa process CPU use karega aur kitne time ke liye.

Real Life Example:

School me teacher decide karta hai ki kaun sa student pehle answer dega. Waise hi scheduler decide karta hai ki kaun sa process pehle CPU use kare.

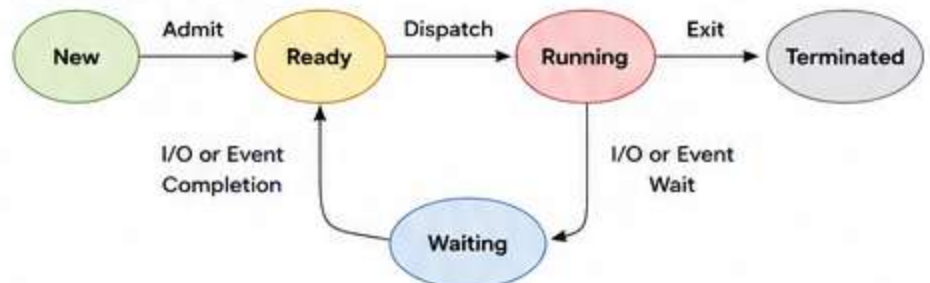
Types of Schedulers

Scheduler	Kaam	Kab Use Hota Hai
Long-term (Job Scheduler)	New jobs ko system me lana	Jab system idle ho
Short-term (CPU Scheduler)	Ready queue se process select karke CPU dena	Bahut baar (milliseconds me)
Medium-term (Swapper)	Processes ko memory se bahar ya andar karna	Jab memory full ho

Short Trick:

L S M = Long, Short, Medium

Process State Diagram



3. OPERATIONS ON PROCESSES

Operation	English Meaning	Hinglish (Simple)	Example
Create	Create a new process	Naya process banate hain	aap Chrome open karte ho
Schedule	Select a process for CPU	Scheduler process select karta hai	OS ready queue se select karta hai
Dispatch	Give CPU to selected process	CPU us process ko de diya jata hai	CPU time start
Block / Wait	Process waits for event	Process kisi event ka wait karta hai	I/O operation, file read
Wakeup	Waking up blocked process	Event complete hone par jagana	Data read ho gaya
Terminate	End the process	Process ka execution khatam	aap app close karte ho

Short Trick:

C S D B W T = Create, Schedule, Dispatch, Block, Wakeup, Terminate

2-Mark Ready Answer:

Operations on processes are: create, schedule, dispatch, block (wait), wakeup and terminate. OS performs these to manage processes effectively.

Long Answer Key Points (12-Mark):

- Need of Operations on Processes
- Detailed explanation of each operation
- State transitions with diagram
- Role of OS in managing these operations

PYQ Indicator:

- 2014 (2M): Which was the first UNIX shell?
- 2016 (2M): What does preemptive mean?
- 2018 (10M): Explain operations on process.

4. THREADS

Definition (English):

A thread is the smallest unit of CPU execution inside a process. A process can have one or more threads.

Hinglish (Simple):

Thread matlab ek chhota worker jo ek process ke andar kaam karta hai. Ek process ke multiple threads ho sakte hain.

Real Life Example:

Ek office me manager (process) ke niche multiple employees (threads) kaam karte hain.

Long Answer Key Points (12-Mark):

- Definition of Thread
- Need of Threads
- Types of Threads
- Benefits and Drawbacks
- Difference between Process and Thread (Table)

Process vs Thread		
Basis	Process	Thread
Definition	Program in execution	Smallest unit of execution inside a process
Memory	Separate memory	Shares memory of process
Resources	Own resources	Shares resources
Creation Time	More time	Less time
Communication	IPC needed	Easy (shared memory)
Context Switch	Slow	Fast
Example	Chrome, VS Code	Multiple tabs of Chrome

PYQ Indicator:

- 2018 (7M): Write short notes on Multithreading.
- 2020 (10M): Explain threads and its types.

Short Trick:

M S R T C C E = Memory, Share, Resources, Time, Communication, Context switch, Example

5. INTER-PROCESS COMMUNICATION (IPC)

Definition (English):

IPC is a mechanism that allows processes to communicate and synchronize with each other.

Hinglish (Simple):

IPC ka matlab hai ki do alag processes ek-dusre se baat kar sake.

Real Life Example:

Do apps (e.g., WhatsApp aur Camera) data share karte hain, ye IPC hai.

Methods of IPC

- Shared Memory
- Message Passing
- Pipes
- Message Queues
- Semaphores
- Sockets

2-Mark Ready Answer:

IPC allows processes to communicate and synchronize. Methods are shared memory, message passing, pipes, semaphores, etc.

PYQ Indicator:

- 2016 (2M): How do you set environment variable which will be accessible from sub shell?
- 2016 (10M): Define IPC and describe any two methods.

6. PRECEDENCE GRAPHS

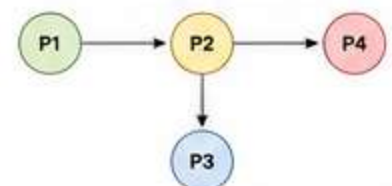
Definition (English):

Precedence graph is a directed graph used to represent the ordering constraints among processes.

Hinglish (Simple):

Ye ek directed graph hota hai jo batata hai ki kaun sa process pehle hoga aur kaun sa baad me.

Diagram (Example)



Real Life Example:

Cooking me pehle chawal pakta hai, phir dal. Ye order batane ke liye precedence graph use karte hain.

2-Mark Ready Answer:

Precedence graph is a directed graph showing ordering of processes. An edge $P_i \rightarrow P_j$ means P_i must finish before P_j starts.

PYQ Indicator:

- 2018 (7M): Write short note on Precedence graph.

Unit One Line Summary:

Process management me OS processes aur threads ko create, schedule, communicate aur synchronize karta hai taa ki system efficiently kaam kare aur resources ka sahi use ho.



FOCUS MORE ON
DIAGRAMS & PYQ

BCA-205 OPERATING SYSTEM & UNIX

UNIT 3: MEMORY MANAGEMENT – CHEAT SHEET

READY FOR
9 JULY TEST

1. CONCEPTS OF MEMORY MANAGEMENT

Definition (English):

Memory Management is the technique of managing primary memory (main memory) in a computer system. It keeps track of which part of memory is being used by whom, allocates memory to processes and deallocates when no longer needed.

Hinglish (Simple):

OS ka kaam hota hai ki kaun sa process kitna memory use karega, kahan store hoga aur kaise manage hoga ye decide karna.

Real Life Example:

Jaise library me books ko rack me rakhna aur jab chahiye nikalna. Waise hi OS memory ko manage karta hai.

PYQ Indicator:

- 2014 (2M): Differentiate between page and segment.
- 2016 (2M): What is meant by contiguous allocation?
- 2018 (10M): Explain paging techniques.
- 2020 (10M): Explain memory management techniques.

2-Mark Ready Answer:

Memory management OS ka ek important function hai jo main memory ko efficiently use karta hai. Yeh memory allocate, deallocate aur protect karta hai.

Long Answer Key Points (12-Mark):

- Need and importance of memory management
- Logical and physical address space
- Allocation methods
- Techniques: Paging, Segmentation, Virtual Memory
- Page replacement algorithms
- Thrashing and its solution
- Protection and security

Short Trick Box

P A V D T S P
= Paging, Allocation, Virtual memory, Demand paging, Thrashing, Security, Protection

2. LOGICAL AND PHYSICAL ADDRESS SPACE

- Logical Address Space: Wo address jo CPU/program generate karta hai. (User view)
- Physical Address Space: Wo actual address jo main memory me exist karta hai. (Hardware view)
- OS logical address ko physical address me map karta hai (via MMU).



3. SWAPPING

Definition (English):

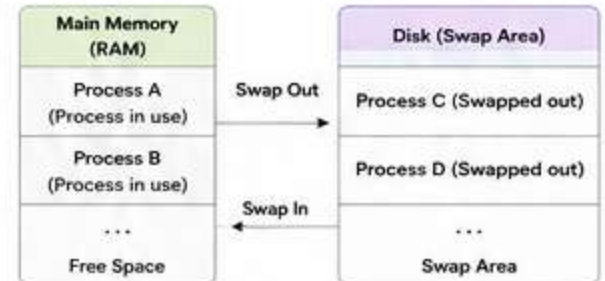
Swapping is a technique of temporarily removing inactive processes from main memory to secondary storage (disk) and bringing them back when needed.

Hinglish (Simple):

Jab memory kam pad jati hai to OS kisi process ko disk (me swap area) me bhej deta hai aur jab zarurat hoti hai usko wapas memory me laata hai.

Real Life Example:

Jaise mobile me app ko background me send kar dete hain taki RAM free ho jaye, waise hi swapping.



PYQ Indicator:

- 2016 (2M): — is a technique of temporarily removing inactive programs from memory.
- 2018 (10M): Explain swapping.

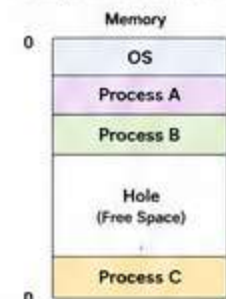
2-Mark Ready Answer:

Swapping is a technique of moving a process from main memory to disk (swap area) to free memory and bring it back when needed.

4. CONTIGUOUS AND NON-CONTIGUOUS ALLOCATION

(A) Contiguous Allocation

- Process ko memory me continuous blocks me allocate kiya jata hai.
- Simple hai but external fragmentation hota hai.



Types:

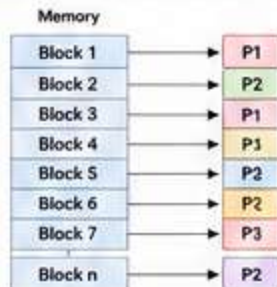
- Fixed Partition
- Variable Partition

Real Life Example:

Jaise hall me logon ko continuous seats me bithana.

(B) Non-Contiguous Allocation

- Process ko memory me non-continuous blocks (frames/blocks) me allocate kiya jata hai.
- External fragmentation nahi hota.



Types:

- Paging
- Segmentation
- Segmentation with Paging

Real Life Example:

Jaise school me books ko alag-alag racks me rakhna, continuous hona zaruri nahi.

Comparison Table

Basis	Contiguous Allocation	Non-Contiguous Allocation
Memory Blocks	Continuous	Non-Continuous
Fragmentation	External Frag. hota hai	External Frag. nahi hota (Only internal ho sakta hai)
Implementation	Simple	Complex
Examples	Fixed, Variable Partition	Paging, Segmentation

Short Trick:

C & N = Contiguous (Continuous)
N-C = Non-Contiguous (No continuous)

PYQ Indicator:

- 2016 (2M): What is meant by contiguous allocation?

5. PAGING

Definition (English):

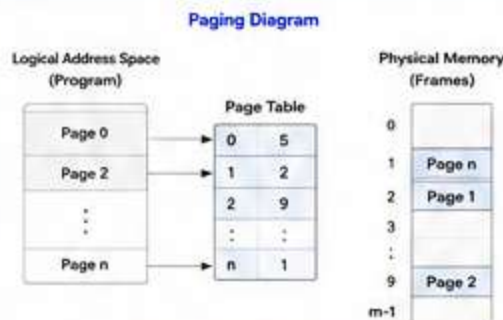
Paging is a memory management technique in which logical memory is divided into fixed-size pages and physical memory is divided into fixed-size frames. Pages of a process can be placed in any free frames.

Hinglish (Simple):

Program ko fixed size ke pages me tod diya jata hai aur memory ko frames me. Page kisi bhi frame me ja sakta hai.

Real Life Example:

Jaise kitab ke pages ko alag-alag box (frames) me rakh dena.



2-Mark Ready Answer:

Paging is a technique in which logical memory is divided into pages and physical memory into frames of same size. Pages are mapped to frames using page table.

Long Answer Key Points (12-Mark):

- Need for paging
- Basic paging mechanism with diagram
- Address translation (page number + offset)
- Page table and its entries
- Advantages and disadvantages

6. SEGMENTATION

Definition (English):

Segmentation is a memory management technique in which a program is divided into logical segments (variable size) such as code, data, stack, etc.

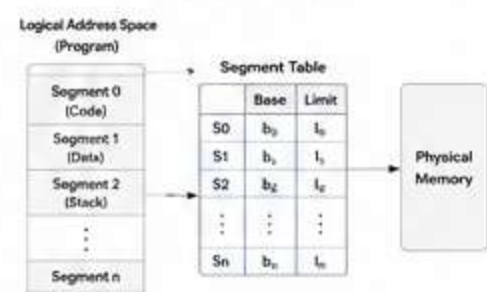
Hinglish (Simple):

Program ko logical parts (segment) me divide karte hain jaise code segment, data segment.

Real Life Example:

Jaise ek file me chapters (segments) hote hain, har chapter alag length ha ho sakta hai.

Segmentation Diagram



2-Mark Ready Answer:

Segmentation divides a program into variable sized logical segments. Each segment has base and limit.

Long Answer Key Points (12-Mark):

- Need for segmentation
- Segment table (base, limit)
- Address translation
- Advantages and disadvantages

7. VIRTUAL MEMORY

Definition (English):

Virtual memory is a memory management technique that allows execution of processes that are partially in main memory and partially on disk.

Hinglish (Simple):

Virtual memory se bada program bhi run ho sakta hai chahe poori RAM me na ho. Sirf required part RAM me aata hai.

Real Life Example:

Jaise apke mobile me kuch apps SD card me hote hain aur jab chahiye use load karte ho.

2-Mark Ready Answer:

Virtual memory is a technique that gives an illusion of large primary memory using secondary storage.

Long Answer Key Points (12-Mark):

- Need for virtual memory
- Demand paging
- Page fault and its handling
- Advantages and disadvantages

8. DEMAND PAGING

Definition (English):

Demand paging is a virtual memory technique in which pages are loaded into memory only when they are demanded (needed) by the process.

Hinglish (Simple):

Page ko tabhi RAM me laaya jata hai jab process us page ko access karta hai.

Real Life Example:

Jaise online video dekhte waqt sirf current part hi load hota hai, pura video nahi.

2-Mark Ready Answer:

Demand paging loads a page into memory only when it is referenced by a process.

9. PAGE REPLACEMENT ALGORITHMS

Common Algorithms

Algorithm	How It Works (Simple)
FIFO (First In First Out)	Jo page sabse pehle aaya tha, sabse pehle remove hoga.
Optimal	Jo page sabse long time tak use nahi hoga, usse replace karte hain. (Best, but not practical)
LRU (Least Recently Used)	Jo page sabse zyada time se use nahi hua, usse replace karte hain.
LFU (Least Frequently Used)	Jo page sabse kam baar use hua hai, usse replace karte hain.

PYQ Indicator:

- 2016 (10M): Explain page replacement algorithms.

Short Trick:

F O L L =
FIFO, Optimal, LRU, LFU

10. ALLOCATION OF FRAMES

- Total number of frames = Total Memory / Frame Size
- Frame allocation can be: Fixed / Variable
- More frames → Less page fault
- But too many frames → Less degree of multiprogramming

11. THRASHING

Definition (English):

Thrashing occurs when system spends more time in paging (swapping pages) than in executing the processes.

Hinglish (Simple):

Jab page fault bahut zyada ho jata hai aur CPU ka zyada time page laane-le jaane me nikal jata hai, useful kaam kam hota hai.

Solutions:

- Increase number of frames
- Working Set Model
- Page Fault Frequency (PFF) algorithm

PYQ Indicator:

- 2016 (10M): What advantage is there in having different time-quantum sizes on different levels of a multilevel queuing system? (Related to Thrashing – OS concept)

12. DEMAND SEGMENTATION

- Combines advantages of demand paging and segmentation.
- Only required segments are brought into memory.

13. SECURITY THREADS PROTECTION, INTRUDERS, VIRUS, TRUSTED SYSTEM

Security	Threads Protection	Intruders	Viruses	Trusted System
OS protects memory using base-limit register, access control, encryption.	Each thread has separate stack and registers. OS ensures no interference.	Unauthorized users trying to gain access to system resources.	Malicious programs that can corrupt or destroy data.	A system that ensures integrity, authentication and confidentiality.

BCA-205 OPERATING SYSTEM & UNIX

UNIT 4: LINUX COMMANDS, vi EDITOR & SHELL - CHEAT SHEET

READY FOR
9 JULY TEST

1. INTRODUCTION TO OPEN SOURCE TECHNOLOGY

Definition (English):

Open Source technology means software whose source code is freely available to everyone. Anyone can use, study, modify and distribute it. Linux is the best example of Open Source OS.

Hinglish (Simple):

Open Source ka matlab hota hai software jiska source code sabke liye khula hota hai. Koi bhi use free me use, change ya share kar sakta hai. Linux open source OS hai.

Real Life Example:

Jaise WhatsApp open API deta hai developers ko apps banane ke liye. Waise hi Linux ka code sabke liye open hai.

PYQ Indicator:

- 2016 (2M): What is meant by Open Source?
- 2020 (2M): Write advantages of Open Source software.

2-Mark Ready Answer:

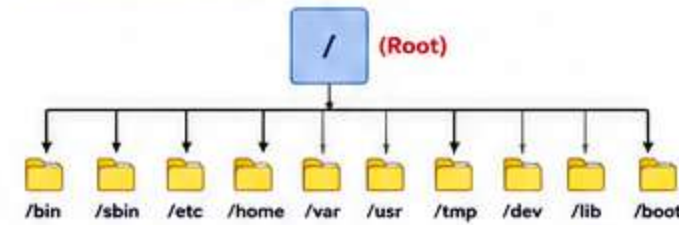
Open Source software ka source code public hota hai. Koi bhi ise use, modify aur distribute kar sakta hai. Linux is a popular open source operating system.

Short Trick:

OPEN = Open, Public, Everyone, No Cost

2. FILE SYSTEM HIERARCHY IN LINUX

Linux File System starts from Root "/"



Directory	Description
/	Root directory of entire file system
/bin	Essential user command binaries (ls, cp, mv, cat ...)
/sbin	System binaries (for system admin)
/etc	Configuration files
/home	User home directories
/var	Variable files (logs, mail, spool, cache)
/usr	User programs, libraries, documentation
/tmp	Temporary files
/dev	Device files (hard disks, cd, pen drive etc.)
/lib	Shared libraries
/boot	Boot loader files, kernel

Real Life Example:

Jaise ghar me alag-alag almari hoti hai - kapde, books, documents ke liye. Waise hi Linux me alag folders hote hai.

PYQ Indicator:

- 2016 (2M): Where should the disk directories be placed?
- 2018 (2M): Explain file system hierarchy in UNIX.

Short Trick:

B S E H V U T D L B => Bin, Sbin, Etc, Home, Var, Usr, Tmp, Dev, Lib, Boot

3. LOGGING IN

Steps:

1. System power on → login prompt appear
 2. Type username and press Enter
 3. Type password (nothing shows on screen)
 4. Successful login → shell prompt appears
- Example: [user@host ~]\$

Real Life Example:

Jaise school gate pe ID card dikha ke entry milti hai. Waise hi Linux me username aur password se login hota hai.

PYQ Indicator:

- 2014 (2M): Describe about logging in.

4. SIMPLE COMMANDS (WITH USE)

Command	Use	Example
ls	List files & directories	ls -l
cp	Copy files/directories	cp a.txt b.txt
mv	Move / Rename files	mv old.txt new.txt
wc	Count lines, words, chars	wc file.txt
sort	Sort lines in file	sort file.txt
cat	Display file content	cat file.txt
grep	Search pattern in file	grep 'word' file.txt
head	Show first 10 lines	head file.txt
tail	Show last 10 lines	tail file.txt
echo	Display a line	echo Hello
touch	Create empty file	touch file.txt
which	Find location of command	which ls
whereis	Find binary, source, man	whereis gcc
whatis	Short description	whatis ls
whoami	Show current user	whoami

Short Trick:

L C M W S C G H T E T W W W W → List, Copy, Move, Word Count, Sort, Cat, Grep, Head, Tail, Echo, Touch, Which, Whereis, Whatis, Whoami

5. DIRECTORY COMMANDS

Command	Use	Example
pwd	Print current directory	pwd
cd	Change directory	cd /home/user
ls	List files/directories	ls -l
mkdir	Make new directory	mkdir newdir
rmdir	Remove empty directory	rmdir olddir
rm	Remove file	rm file.txt
rm -r	Remove directory & contents	rm -r mydir

Real Life Example:

Jaise ghar me room change karna (cd), naya room banana (mkdir), kabada hatana (rm).

PYQ Indicator:

- 2014 (2M): To remove a directory - which command is used?
- 2016 (2M): Write syntax of mkdir command.

2-Mark Ready Answer:

Directory banane ke liye "mkdir" command use hota hai. Directory hatane ke liye "rmdir" ya "rm -r" use hota hai.

Short Trick:

P C L M R R R → Print path, Change dir, List, Make dir, Remove empty dir, Remove file, Remove dir recursively

6. PERMISSION COMMANDS

(A) chmod - Change permission

Permission	User (Owner)	Group	Others
Read (r)	4	4	4
Write (w)	2	2	2
Execute (x)	1	1	1

Example:

- 777 = rwx rwx rwx (Read+Write+Execute for all)
- 755 = rwx r-x r-x (Owner full, others read+execute)
- 644 = rw- r-- r-- (Owner read+write, others read only)

Syntax: chmod [options] mode file

(B) chown - Change owner

Syntax: chown user file

Example: chown john file.txt

(C) chgrp - Change group

Syntax: chgrp group file

Example: chgrp students file.txt

PYQ Indicator:

- 2014 (2M): The octal number used with chmod command to make a file read, write, execute to owner is?
- 2018 (2M): Explain permission commands.

Short Trick:

R=4, W=2, X=1 → Add numbers and set permission

7. vi EDITOR

vi is a screen oriented text editor.

(A) Modes in vi

Mode	Use
Command Mode	Default mode, for navigation & commands
Insert Mode	For typing text
Last Line Mode	For saving, quit, search, replace

(B) How to Switch Modes

- Command → Insert : press "i" (insert)
- Insert → Command : press "Esc"
- Command → Last Line : press ":" (colon)

(C) Important vi Commands

Command	Use
i	Enter insert mode (before cursor)
a	Enter insert mode (after cursor)
x	Delete character under cursor
dd	Delete current line
yy	Yank (copy) current line
p	Paste yanked line
:w	Write (save) file
:q	Quit vi
:wq	Save and quit
:q!	Quit without saving
/text	Search forward
?text	Search backward

2-Mark Ready Answer:

vi editor ke 3 modes hote hai - Command, Insert, Last Line. Mode change i, Esc, : keys se hota hai.

Short Trick:

i = insert
Esc = back to command mode
: = last line mode

PYQ Indicator:

- 2014 (2M): To erase the entire word in vi editor which key is used? Answer: Ctrl-w
- 2016 (2M): What command is used to replace many characters in vi editor? Answer: r (replace) or :s/old/new/

8. SHELL

Definition (English):

Shell is a command line interpreter that acts as interface between user and OS kernel. It reads commands from user and gives output.

Hinglish (Simple):

Shell wo program hai jo user ke command ko samajhta hai aur OS ko execute karne ke liye bhejta hai. Jo output aata hai, wo user ko dikhata hai.

Types of Shell:

- Bourne Shell (sh) - First UNIX shell
- C Shell (csh)
- Korn Shell (ksh)
- Bash Shell (bash) - Most popular in Linux

PYQ Indicator:

- 2014 (2M): Which was the first UNIX shell? Answer: Bourne shell
- 2016 (2M): How many types of shell are used in UNIX?

2-Mark Ready Answer:

Shell user aur kernel ke beech interface ka kaam karta hai. Bourne shell pehla UNIX shell tha.

9. I/O REDIRECTION

I/O Redirection ka matlab hai input ya output ko file ya kisi other source/destination me bhejna.

Types of Redirection

Symbol	Name	Use	Example
>	Output Redirection	Output ko file me likhna (overwrite)	command > file.txt
>>	Append Redirection	Output ko file me add karna (append)	command >> file.txt
<	Input Redirection	Input file se lena	command < file.txt
2>	Error Redirection	Error output ko file me lena	command 2> err.txt
&>	Output + Error	Output aur error dono file me	command &> out.txt

Real Life Example:

Jaise printer ka output paper pe jata hai. Redirection se output file me save hota hai.

PYQ Indicator:

- 2016 (2M): What is I/O redirection?
- 2020 (10M): Explain I/O redirection and piping commands.

Short Trick:

> Overwrite, >> Append, < Input, 2> Error, &> All

10. PIPING (|)

Piping ka use ek command ka output dusre command ke input ke roop me dene ke liye hota hai.

Syntax:

command1 | command2

Example:

- ls -l | more (long output page by page)
- cat file.txt | grep 'word' (search word in file)
- ps -ef | grep 'java' (search java process)

Real Life Example:

Jaise ek nal se pani aaya aur dusre pipe me chala gaya bina rukawaat ke.

PYQ Indicator:

- 2016 (2M): Define pipe and describe any two methods.

2-Mark Ready Answer:

Pipe (|) ek command ka output dusre command ke input me deta hai. Isse data flow continuous hota hai.

Short Trick:

| = Pipe (Next command ka input banata hai)

UNIT 4 ONE LINE SUMMARY

Linux me files ko manage karne, commands chalane, permissions set karne, text edit karne (vi), commands ko execute/synchronize karne (shell, redirection, piping) ke tools provide hote hai.

MOST IMPORTANT COMMANDS (QUICK LIST)

ls, cd, pwd, mkdir, rmdir, rm, cp, mv, cat, more, less, head, tail, wc, sort, grep, find, chmod, chown, chgrp, ps, kill, touch, echo, man, which, whereis, df, du, clear, whoami, date, cal

FOCUS POINTS (UNIT 4)

- ✓ File System Hierarchy yaad karo
- ✓ Basic Commands daily practice karo
- ✓ vi Modes aur commands yaad karo
- ✓ Redirection & Piping examples samjho

PYQ HINTS (UNIT 4)

- Simple commands - 2M
- vi editor modes/commands - 2M
- chmod octal number - 2M
- I/O redirection & piping - 2M

1. FEATURES OF LINUX

- **Open Source:** Linux is free to use, modify and distribute.
- **Multituser:** Many users can work at the same time.
- **Multitasking:** Can run many tasks simultaneously.
- **Portable:** Runs on different hardware (PC, Server, Mobile, Embedded).
- **Secure:** Strong permission system, less virus affected.
- **Stable & Reliable:** Less crash, 24x7 running without hang.
- **High Performance:** Efficient memory and CPU usage.
- **Large Community Support:** Huge online help and documentation.
- **Customizable:** User can change look, feel and functionality.
- **Supports many File systems:** ext4, xfs, btrfs, etc.
- **Powerful Networking:** Supports TCP/IP, SSH, Server services.

Real Life Example:

Jaise Android mobile ka core Linux par based hai. Server, Cloud, Super Computers me Linux ka use hota hai.

PYQ Indicator:

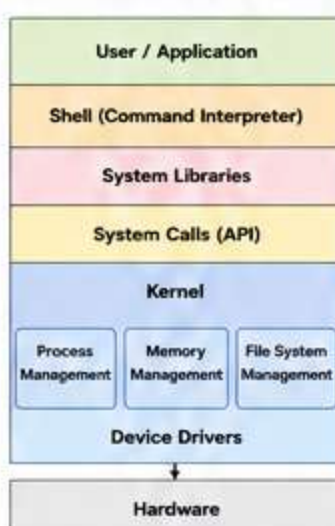
- 2014 (2M): List features of Linux.
- 2016 (2M): Write advantages of Linux.
- 2018 (10M): Explain features of Linux in detail.
- 2020 (10M): Compare Linux with other OS.

Short Trick:

O M P S S R H C P C

O-open source, M-multituser, P-multitasking, S-secure, S-stable, R-reliable, H-high performance, C-community support, P-portable, C-customizable

2. COMPONENTS OF LINUX



Real Life Example:

Jaise school me Principal (Kernel) sab manage karta hai, Librarian (Libraries) books provide karta hai, Clerk (Shell) commands leta hai aur kaam karata hai.

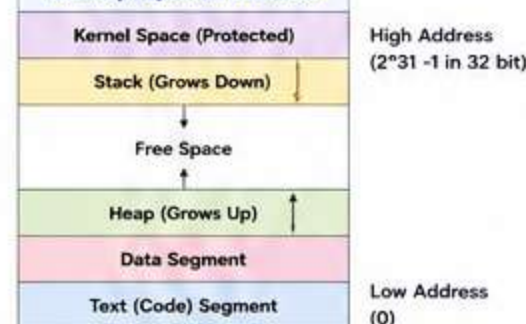
Short Trick:

S K L S D = Shell, Kernel, Libraries, System Calls, Drivers.

3. MEMORY MANAGEMENT SUBSYSTEMS

- Linux uses virtual memory for efficient management.
- Main components:
 1. **Virtual Memory:** Each process ko alag memory space milta hai.
 2. **Paging:** Memory ko fixed size pages me divide ta hai.
 3. **Page Table:** Logical address ko physical address me map karta hai.
 4. **Swapping:** Inactive process ko disk par swap out kiya jata hai.
 5. **Cache & Buffer:** Frequently used data store karke performance badhata hai.

Memory Layout of a Process



PYQ Indicator:

- 2016 (2M): What is swapping?
- 2018 (10M): Explain paging technique.
- 2020 (10M): Describe memory management in Linux.

4. LINUX PROCESS AND THREAD MANAGEMENT

Process Management

- Process is a program in execution.
- Linux provides multitasking using processes.
- Commands:
 - ps - show running processes
 - top - dynamic process view
 - kill - terminate process
 - nice / renice - change priority

Thread Management

- Thread is smallest unit of CPU execution.
- Threads share code, data and resources of process.
- **Benefits:** Fast, less memory, better communication.
- **Commands/Tools:** ps -L, top -H

Process vs Thread

Basis	Process	Thread
Definition	Program in execution	Smallest unit of execution inside a process
Memory	Separate memory space	Shares memory of process
Creation Time	More time	Less time
Communication	IPC needed	Easy (shared memory)
Context Switch	Slow	Fast
Example	Chrome, VS Code	Multiple tabs of Chrome

PYQ Indicator:

- 2018 (7M): Write short note on Threads.
- 2020 (10M): Explain process and thread management in Linux.

5. FILE MANAGEMENT SYSTEM

- Linux treats everything as file.
- Types of Files:
 - Regular File (-)
 - Directory (d)
 - Character Device (c)
 - Block Device (b)
 - Symbolic Link (l)
 - Named Pipe (p)
 - Socket (s)

File Permission (rwx)

r = read (4)
w = write (2)
x = execute (1)

Example: -rwxr-xr--

User (Owner)	Group	Others
rwx (7)	r-x (5)	r-- (4)

Real Life Example:

Jaise ghar ke documents alag-alag folders (directories) me hote hain aur har file par alag permission hota hai.

PYQ Indicator:

- 2014 (2M): List file types in Linux.
- 2016 (2M): Write file access modes.
- 2018 (10M): Explain file system in UNIX.

Useful Commands

Command	Use
ls -l	Long list with permission
cat file	Display file content
less file	View file page by page
cp src dest	Copy file
mv src dest	Move/Rename file
rm file	Remove file
ln -s target link	Create symbolic link
df -h	Disk space free
du -sh dir	Directory size

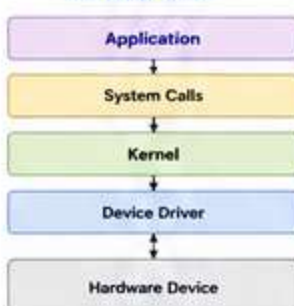
Short Trick:

r=4, w=2, x=1 → rwx = 7
E.g. 755 = rwxr-xr-x

6. DEVICE DRIVERS

- Device Driver is a program that allows OS to communicate with hardware devices.
- Linux has huge collection of drivers for various devices.
- Drivers handle: I/O operations, interrupts, device initialization.
- Commands:
 - lsmod - list loaded modules
 - modprobe name - load module
 - rmmod name - remove module

How Driver Works



Real Life Example:

Jaise printer ko computer se chalane ke liye driver install karte hain, warna OS use communicate nahi kar sakta.

PYQ Indicator:

- 2016 (10M): Explain device drivers in Linux.

Short Trick:

D R I V E R = Device ko Recognize kare, Interface provide kare, Verify kare, Execute request kare, Response de

7. UNIX SYSTEM ADMINISTRATION

- System Administrator (root/super user) system ko setup, manage, secure aur maintain karta hai.
- Main tasks:
 1. User Management
 2. File System Management
 3. System Monitoring
 4. Backup and Restore
 5. Security Management
 6. Software Installation



System Administrator

8. INSTALLING LINUX

General Steps:

1. Choose Distribution (Ubuntu, Fedora, Debian etc.)
2. Download ISO File
3. Create Bootable USB/DVD
4. Boot from USB/DVD
5. Start Installer
6. Partition Disk
7. Set User & Password
8. Install & Reboot

Real Life Example:

Jaise naya software install karte hain waise hi Linux OS install karte hain par partition aur user settings ke saath.

PYQ Indicator:

- 2020 (10M): Describe steps of installing Linux.

9. USER ACCOUNTS

- Linux is multiuser system.
- Each user has username, UID, group, home directory, shell.
- Types of Users:
 1. **Super User (root)** - full control
 2. **Normal User** - limited rights

Useful Commands

Command	Use
adduser name	Add new user
userdel name	Delete user
passwd name	Set/Change password
id name	Show user info
groups name	Show groups of user
who	Logged in users
w	Who is what

PYQ Indicator:

- 2014 (2M): Write commands to add and delete user.
- 2018 (10M): Explain user management in UNIX.

Real Life Example:

Jaise office me alag-alag staff ka ID aur access level hote hain, waise Linux me users aur groups hote hain.

Short Trick:

A U P I G W W = Add user, Userdel, Passwd, ID, Groups, Who, W

10. BACKUPS AND RESTORATION

Backup Commands

Command	Use
tar -cvf file.tar dir	Create archive
tar -xvf file.tar	Extract archive
cp -r dir /backup	Copy directory
rsync -av src dest	Sync and backup

Restore Example

```
tar -xvf backup.tar
rsync -av /backup/data /data
```



Real Life Example:

Jaise mobile ka data cloud me backup rakhne hain taki data loss na ho, waise hi Linux me backup lete hain.

PYQ Indicator:

- 2016 (2M): What is backup?
- 2020 (10M): Explain backups and restoration in UNIX.

Short Trick:

B A C K U P = Before Any Crash Keep Useful Precaution

QUICK REVISION (UNIT 5)

- ✓ Linux is Open Source, Multituser, Multitasking.
- ✓ Kernel is heart of Linux.
- ✓ Everything is a file in Linux.
- ✓ Use ls -l to see permission and details.
- ✓ r=4, w=2, x=1 (chmod numbers).
- ✓ Processes are heavy, threads are light.
- ✓ Use ps, top, kill for process management.
- ✓ Swap is used when memory is full.
- ✓ Root user has full control.
- ✓ Backup important data using tar/rsync.

UNIT 5 ONE LINE SUMMARY

Linux powerful features, components, memory, process, file system, drivers and system administration tools ke through ek secure, stable aur multiuser environment provide karta hai.

MOST IMPORTANT COMMANDS (UNIT 5)

ls, ls -l, cat, cp, mv, rm, mkdir, rmdir, chmod, chown, ps, top, kill, adduser, passwd, userdel, df -h, du -sh, tar -cvf, tar -xvf, rsync -av, who, w, id

FOCUS POINTS

- ★ Features & Components of Linux
- ★ File system & Permissions
- ★ Process & Thread Management
- ★ User Management & Backup

PYQ HINTS (UNIT 5)

- 2M → Short definitions, lists, features
- 7M → Short notes (threads, drivers)
- 10M → Detailed explanation (install, file system, backup, security, admin tasks)

★★★★★ VERY IMPORTANT (12 MARKS) QUESTIONS

UNIT 1 INTRODUCTION TO OPERATING SYSTEMS

1. Explain Operating System and its Services.
2. Explain CPU Scheduling Algorithms (FCFS, SJF, Priority, Round Robin).
3. Difference between Multiprogramming and Time-sharing.
4. Explain System Calls with types.
5. Explain Scheduling Criteria.

Most Repeated PY Years
2014, 2016, 2018, 2020, 2022, 2024

UNIT 2 PROCESS MANAGEMENT

1. Explain Process and Process State Diagram.
2. Explain Threads (User, Kernel, Hybrid).
3. Explain Critical Section Problem.
4. Explain Semaphore with Producer-Consumer Problem.
5. Explain Deadlock, conditions and Banker's Algorithm.

Most Repeated PY Years
2014, 2015, 2017, 2019, 2021, 2023, 2025

UNIT 3 MEMORY MANAGEMENT

1. Explain Paging with page table.
2. Explain Page Replacement Algorithms (FIFO, LRU, Optimal).
3. Explain Paging vs Segmentation.
4. Explain Virtual Memory & Demand Paging.
5. Explain Thrashing and its solution.

Most Repeated PY Years
2014, 2016, 2017, 2019, 2022, 2024, 2025

UNIT 4 LINUX & UNIX

1. Explain Linux File System Hierarchy.
2. Explain vi Editor Modes and Commands.
3. Explain Shell and Types of Shell.
4. Explain Linux Commands (ls, cp, mv, cat, grep, wc, chmod, chown, etc.).
5. Explain I/O Redirection and Piping.

Most Repeated PY Years
2014, 2015, 2016, 2018, 2020, 2022, 2024, 2025

UNIT 5 FEATURES OF LINUX & UNIX SYSTEM ADMINISTRATION

1. Explain Features of Linux.
2. Explain Components of Linux.
3. Explain Linux Process & Thread Management.
4. Explain Role of UNIX System Administrator.
5. Explain Backup and Restoration.

Most Repeated PY Years
2015, 2017, 2018, 2020, 2021, 2023, 2025

★ FREQUENTLY ASKED (2-5 MARKS)

- What is Operating System?
- Define Process.
- Define Thread.
- What is Scheduler?
- What is Multiprogramming?
- What is Time-sharing?
- What is System Call?
- What is Real-Time System?
- What is I/O Buffering?
- What is Device Driver?

- Define Paging.
- Define Segmentation.
- What is Virtual Memory?
- What is Demand Paging?
- What is Page Replacement?
- What is Page Fault?
- What is Thrashing?
- What is Swapping?
- What is Deadlock?
- What is Semaphore?

- What is Critical Section?
- What is Shell?
- What are Types of Shell?
- What is vi Editor?
- What are vi Modes?
- What is File Permission?
- What is I/O Redirection?
- What is Piping?
- What is Backup?
- What is UNIX?

Topic	Importance	Asked in Years
CPU Scheduling Algorithms	★★★★★	2014,16,17,18,20,22,24,25
Paging	★★★★★	2014,16,17,19,22,24,25
Deadlock & Banker's Algorithm	★★★★★	2014,15,17,19,21,23,25
Linux Commands	★★★★★	2014,15,16,18,20,22,24,25
Shell & Types of Shell	★★★★☆	2014,15,16,18,20,22,24
vi Editor	★★★★☆	2014,16,18,20,22,24
Process & Threads	★★★★☆	2014,15,17,19,21,23,25
Linux Features	★★★★☆	2014,16,18,20,21,23,25
System Calls	★★★★☆	2014,15,17,19,21,24
UNIX System Admin	★★★☆☆	2015,17,18,20,23,25

UNIT-WISE HIGH FREQUENCY TOPICS SUMMARY

UNIT 1 INTRODUCTION TO OS

- OS Definition & Services
- Types of OS
- System Components
- System Calls (Types)
- CPU Scheduling Algorithms
- Scheduling Criteria
- Multiprogramming
- Time-sharing
- Real-Time Systems
- I/O Management & Buffering

UNIT 2 PROCESS MANAGEMENT

- Process Concept
- Process State Diagram
- Process Control Block (PCB)
- CPU Scheduling Algorithms
- Threads: User, Kernel, Hybrid
- Multithreading Models
- Critical Section Problem
- Synchronization
- Semaphores (Types)
- Producer-Consumer Problem
- Deadlock (Conditions)
- Deadlock Handling
- Banker's Algorithm

UNIT 3 MEMORY MANAGEMENT

- Memory Management
- Address Binding
- Logical vs Physical Address
- Paging (with example)
- Page Table
- Page Replacement Algorithms
 - FIFO
 - LRU
 - Optimal
- Paging vs Segmentation
- Segmentation
- Virtual Memory
- Demand Paging
- Thrashing & Solution

UNIT 4 LINUX & UNIX

- Linux File System Hierarchy
- Important Linux Commands
 - File & Directory
 - Text Processing
 - Others
- vi Editor (Modes & Commands)
- Shell (Types)
- Shell Scripting (Basics)
- I/O Redirection
- Piping
- File Permissions (rwx)
- chmod, chown, chgrp

UNIT 5 FEATURES & ADMINISTRATION

- Features of Linux
- Components of Linux
- Linux Process Management
- Linux Thread Management
- System Monitoring Tools
- Device Drivers
- UNIX System Administrator
 - Roles & Responsibilities
 - User Management
 - File & Permission Mgmt.
 - Backup & Restoration
 - System Maintenance
 - Security Management

✦ MUST LEARN DIAGRAMS & TABLES

- ✓ Process State Diagram
- ✓ Thread Models (User, Kernel, Hybrid)
- ✓ CPU Scheduling Gantt Charts
- ✓ Paging Diagram with Page Table
- ✓ Logical vs Physical Address Diagram
- ✓ Page Replacement Steps (FIFO, LRU, Optimal)
- ✓ Deadlock Resource Allocation Graph
- ✓ Banker's Algorithm Safe State Table
- ✓ Linux File System Hierarchy Tree
- ✓ vi Editor Modes Diagram
- ✓ I/O Redirection & Piping Diagrams

✓ EXAM TIPS

1. Answer theory questions with definitions, diagrams and examples.
2. For algorithms, write steps + example + table.
3. Use diagrams for Process State, Paging, Deadlock.
4. For Linux commands, write syntax + example.
5. Practice vi editor commands in order.
6. Write points in bullet form for 2-5 marks.
7. Time Management: 1 mark = 2-2.5 minutes.

🕒 LAST 3 HOURS REVISION PLAN

1. CPU Scheduling (1 hr)
2. Process & Deadlock (45 min)
3. Paging & Page Replacement (45 min)
4. Virtual Memory & Thrashing (30 min)
5. Linux Commands (45 min)
6. Shell & vi Editor (30 min)
7. Linux Features & Components (20 min)
8. UNIX Administration (25 min)
9. Diagrams & Tables Quick Revision (20 min)



100 IMPORTANT LINUX COMMANDS CHEAT SHEET

BCA-205 Operating System & UNIX ★ Quick Revision ★ 9 July Test Special

Remember:
Linux is Case Sensitive
Use TAB for Auto Complete

1. FILE & DIRECTORY COMMANDS (1–20)

No.	Command	Use
1	ls	List files & directories
2	ls -l	List with long format
3	ls -a	List all files (including hidden)
4	ls -lh	Human readable format
5	pwd	Print working directory
6	cd dir	Change directory
7	cd ..	Go to parent directory
8	cd ~	Go to home directory
9	mkdir dir	Make new directory
10	rmdir dir	Remove empty directory
11	rm file	Remove file
12	rm -r dir	Remove directory recursively
13	rm -f file	Force remove file
14	cp src dest	Copy file
15	cp -r src dest	Copy directory recursively
16	mv src dest	Move/rename file or directory
17	touch file	Create empty file
18	cat file	Display file content
19	more file	View file page by page
20	less file	View file with scrolling

2. FILE CONTENT & TEXT COMMANDS (21–40)

No.	Command	Use
21	head file	Show first 10 lines
22	head -n 20 file	Show first 20 lines
23	tail file	Show last 10 lines
24	tail -n 20 file	Show last 20 lines
25	tail -f file	Monitor file real time
26	wc file	Count lines, words, bytes
27	wc -l file	Count lines
28	wc -w file	Count words
29	wc -c file	Count bytes
30	sort file	Sort lines
31	sort -r file	Sort in reverse order
32	uniq file	Remove duplicate lines
33	grep pattern file	Search pattern in file
34	grep -i pattern file	Case insensitive search
35	grep -r pattern dir	Search in directory recursively
36	cut -d: -f1 file	Cut columns (example)
37	tr 'a-z' 'A-Z' < file	Convert lower to upper case
38	sed 's/old/new/' file	Replace text in file
39	awk '{print \$1}' file	Print first column
40	echo text	Display text

3. PERMISSION & OWNERSHIP (41–60)

No.	Command	Use
41	chmod 755 file	Change file permission
42	chmod 644 file	Read/Write for owner
43	chmod +x file	Add execute permission
44	chmod -x file	Remove execute permission
45	chmod u+r file	Add read for owner
46	chmod g+w file	Add write for group
47	chmod o-x file	Remove execute for others
48	chown user file	Change owner
49	chown user:group file	Change owner & group
50	chgrp group file	Change group
51	umask 022	Set default permission
52	ls -l	Check permissions
53	stat file	Show file status
54	id	Show user and group id
55	whoami	Show current user
56	groups	Show user groups
57	sudo command	Execute as superuser
58	sudo -i	Switch to root user
59	su user	Switch to another user
60	passwd	Change password

4. PROCESS MANAGEMENT (61–80)

No.	Command	Use
61	ps	Show running processes
62	ps aux	Show all processes
63	ps -ef	Full format process list
64	top	Real-time process monitor
65	htop	Interactive process viewer
66	kill pid	Kill process by PID
67	kill -9 pid	Force kill process
68	pkill name	Kill process by name
69	pgrep name	Get PID by process name
70	nice -n 10 cmd	Run with nice value
71	renice 5 -p pid	Change nice value
72	jobs	Show background jobs
73	bg job	Resume job in background
74	fg job	Bring job to foreground
75	nohup cmd &	Run command immune to hangup
76	time cmd	Measure execution time
77	uptime	Show system uptime
78	w	Who is logged in
79	who	Show logged in users
80	last	Show last logged in users

5. SYSTEM INFORMATION (81–90)

No.	Command	Use
81	uname -a	System information
82	hostname	Show hostname
83	hostname -I	Show IP address
84	date	Show system date
85	cal	Show calendar
86	df -h	Disk space (human readable)
87	df -i	Inode information
88	free -h	Memory usage
89	du -sh dir	Directory size summary
90	du -h dir	Directory size (detailed)

7. NETWORK COMMANDS (96–98)

No.	Command	Use
96	ifconfig / ip addr	Show IP address
97	ping domain.com	Check connectivity
98	netstat -tulnp	Show network connections

8. SERVICE & SYSTEM CONTROL (99–100)

No.	Command	Use
99	systemctl status service	Check service status
100	systemctl start service	Start service

6. COMPRESSION & ARCHIVES (91–95)

No.	Command	Use
91	tar -cvf file.tar dir	Create tar archive
92	tar -xvf file.tar	Extract tar archive
93	tar -czvf file.tar.gz dir	Create tar.gz archive
94	tar -xzvf file.tar.gz	Extract tar.gz archive
95	zip file.zip file	Create zip archive

WILDCARDS (PATTERNS)	
*	Matches any number of characters
?	Matches any single character
[abc]	Matches any one character in brackets
[!abc]	Matches any character not in brackets

REDIRECTION & PIPING	
>	Redirect output (overwrite)
>>	Redirect output (append)
<	Input from file
	Pipe output to another command
2>	Redirect error output

EXAMPLES (Frequently Used)	
• ls -l /home	• tar -czvf file.tar.gz /home/user
• cp file.txt /backup/	• df -h
• mv old.txt new.txt	• free -h
• grep "linux" file.txt	
• tar -czvf backup.tar.gz /home/user	
• ps aux grep apache	
• ps aux sort -nr head -n 10	
• Find apache process	

TIPS TO REMEMBER
✓ Use TAB for auto complete
✓ Use man command for manual
✓ Use history to see previous commands
✓ Use !! to repeat last command
✓ Linux is case sensitive
✓ Practice more = Score more

SPECIAL SHORTCUTS	
Ctrl + C	Stop current command
Ctrl + Z	Suspend current command
Ctrl + D	Logout / Exit
Ctrl + L	Clear screen
Ctrl + R	Search in history

man command
Get manual of any command
Example: man ls



PRACTICE THESE COMMANDS REGULARLY TO BECOME LINUX EXPERT!



ALL THE BEST FOR 9 JULY TEST!



BCA-205 OPERATING SYSTEM & UNIX

LAST NIGHT REVISION SHEET – TOP 50 QUESTIONS

9 JULY TEST
BE READY!



Very Important



Most Repeated PYQs (2014–2025)



High Weightage (5–12 Marks)

UNIT 1: INTRODUCTION TO OPERATING SYSTEMS



1. What is Operating System? Explain its functions. (2)
2. Explain the services provided by Operating System. (5)
3. What is System Call? Explain types of system calls. (5)
4. Explain CPU Scheduling. What are its objectives? (3)
5. Explain the following CPU Scheduling algorithms: FCFS, SJF, Priority, Round Robin. (8)
6. What is Multiprogramming? Explain its advantages. (3)
7. What is Time-sharing system? How is it different from Multiprogramming? (3)
8. Explain I/O Management and its objectives. (2)
9. What is Real-Time System? Give its types. (2)
10. What are the Scheduling Criteria? Explain. (5)

HIGHLY REPEATED TOPICS

- ✓ CPU Scheduling, System Calls, Services, Multiprogramming, Time-sharing, Scheduling Criteria

UNIT 2: PROCESS MANAGEMENT



11. What is a Process? Explain process states. (3)
12. Draw and explain Process State Diagram. (5)
13. What is PCB? What are the contents of PCB? (3)
14. What is Context Switch? (2)
15. What is Thread? Explain types of threads. (4)
16. What is Multithreading? Give its advantages. (3)
17. What is Critical Section Problem? (2)
18. What are the requirements to solve Critical Section Problem? (3)
19. Explain Semaphore. How is it different from Mutex? (4)
20. Explain Producer–Consumer Problem using Semaphore. (5)

HIGHLY REPEATED TOPICS

- ✓ Process State Diagram, Threads, Critical Section, Semaphore, Producer–Consumer

UNIT 3: MEMORY MANAGEMENT



21. What is Memory Management? Why is it needed? (2)
22. Explain Paging with neat diagram. (5)
23. What is Page Table? Explain. (3)
24. Explain Page Replacement Algorithms: FIFO, LRU, Optimal. (8)
25. What is Belady's Anomaly? (2)
26. Explain Paging vs Segmentation. (4)
27. What is Virtual Memory? (3)
28. Explain Demand Paging. (3)
29. What is Thrashing? How can it be controlled? (3)
30. What is Swapping? (2)

HIGHLY REPEATED TOPICS

- ✓ Paging, Page Replacement, Virtual Memory, Demand Paging, Thrashing

UNIT 4: LINUX COMMANDS, vi EDITOR & SHELL



31. Draw and explain Linux File System Hierarchy. (5)
32. What is Shell? Explain its types. (3)
33. What is the use of bash shell? (2)
34. List and explain some important Linux commands with examples: ls, cp, mv, rm, cat, more, less. (5)
35. What is the use of grep, wc, sort, uniq, cut, tr? (4)
36. Explain chmod command with examples. (3)
37. Explain chown and chgrp commands. (3)
38. What is vi editor? Explain its modes. (3)
39. List vi editor commands in each mode. (4)
40. What is I/O Redirection? Explain with examples. (5)

HIGHLY REPEATED TOPICS

- ✓ Linux Commands, vi Editor, Shell, Permissions, I/O Redirection

UNIT 5: FEATURES OF LINUX & UNIX SYSTEM ADMINISTRATION

41. What are the features of Linux? (5)
42. Explain components of Linux. (3)
43. Explain Linux Process Management. (5)
44. Explain Linux Thread Management. (3)
45. What are device drivers? Explain types. (3)
46. What is the role of UNIX System Administrator? (3)
47. What are the responsibilities of System Administrator? (3)
48. Explain user account management in UNIX. (3)
49. Explain backup and restoration in UNIX. (3)
50. What are the commands used for system monitoring? (top, ps, df, du, free, uname) (3)

HIGHLY REPEATED TOPICS

- ✓ Features, Components, Process & Thread Mgmt, UNIX Admin, Backup & Restoration

VERY SHORT QUESTIONS (1–2 MARKS)

1. Define Operating System.
2. Define Process.
3. Define Thread.
4. What is Scheduler?
5. What is PCB?
6. What is Context Switch?
7. What is Multiprogramming?
8. What is Time-sharing?
9. What is System Call?
10. What is a Semaphore?
11. What is Mutex?
12. What is Deadlock?
13. What is Paging?
14. What is Segmentation?
15. What is Virtual Memory?
16. What is Page Fault?
17. What is Swapping?
18. What is Shell?
19. What is vi editor?
20. What is chmod 755?

IMPORTANT DIAGRAMS TO REVISION

- Process State Diagram
- CPU Scheduling Gantt Charts
- Paging Diagram with Page Table
- Logical vs Physical Address
- Page Replacement Steps
- Deadlock Resource Allocation Graph
- vi Editor Modes Diagram
- I/O Redirection & Piping Diagram
- Linux File System Hierarchy Tree

MOST REPEATED PYQS (2014–2025)

- ★ CPU Scheduling Algorithms
- ★ Paging & Page Replacement
- ★ Deadlock & Banker's Algorithm
- ★ Linux Commands & vi Editor
- ★ System Calls & Services
- ★ Shell & Permissions
- ★ Virtual Memory & Thrashing
- ★ UNIX System Administration

FORMULA / SHORT NOTES

- CPU Utilization = $\frac{\text{Busy Time}}{\text{Total Time}}$
- Throughput = $\frac{\text{No. of jobs completed}}{\text{Total Time}}$
- Turnaround Time = Completion Time – Arrival Time
- Waiting Time = Turnaround Time – Burst Time
- Response Time = First Response – Arrival Time
- Page Fault Rate = $\frac{\text{Page Faults}}{\text{Total References}}$
- hit ratio = $1 - \text{Page Fault Rate}$

TIME MANAGEMENT (LAST NIGHT PLAN)

- 1st Hour : Unit 1 (Intro + CPU Scheduling)
- 2nd Hour : Unit 2 (Process Mgmt + Deadlock)
- 3rd Hour : Unit 3 (Memory Mgmt)
- 4th Hour : Unit 4 (Linux + vi + Shell)
- 5th Hour : Unit 5 (Linux Features + UNIX Admin)
- Last 30 Min : Revise Diagrams, Commands, Short Qs



EXAM TIPS

- ✓ Read question carefully.
- ✓ Draw diagrams wherever needed.
- ✓ Write step by step in algorithms.
- ✓ Use examples in 5 or 8 marks answers.
- ✓ Manage time – Easy first, tough later.



SCORE HIGH – REMEMBER

- ✓ Write neat & structured answers.
- ✓ Use bullet points and diagrams.
- ✓ Learn important commands by heart.
- ✓ Practice PYQs and old papers.
- ✓ Stay Calm & Confident!



STAY FOCUSED



PRACTICE PYQs = SECURE MARKS



ALL THE BEST FOR 9 JULY TEST!



9 JULY TEST
Be Ready!

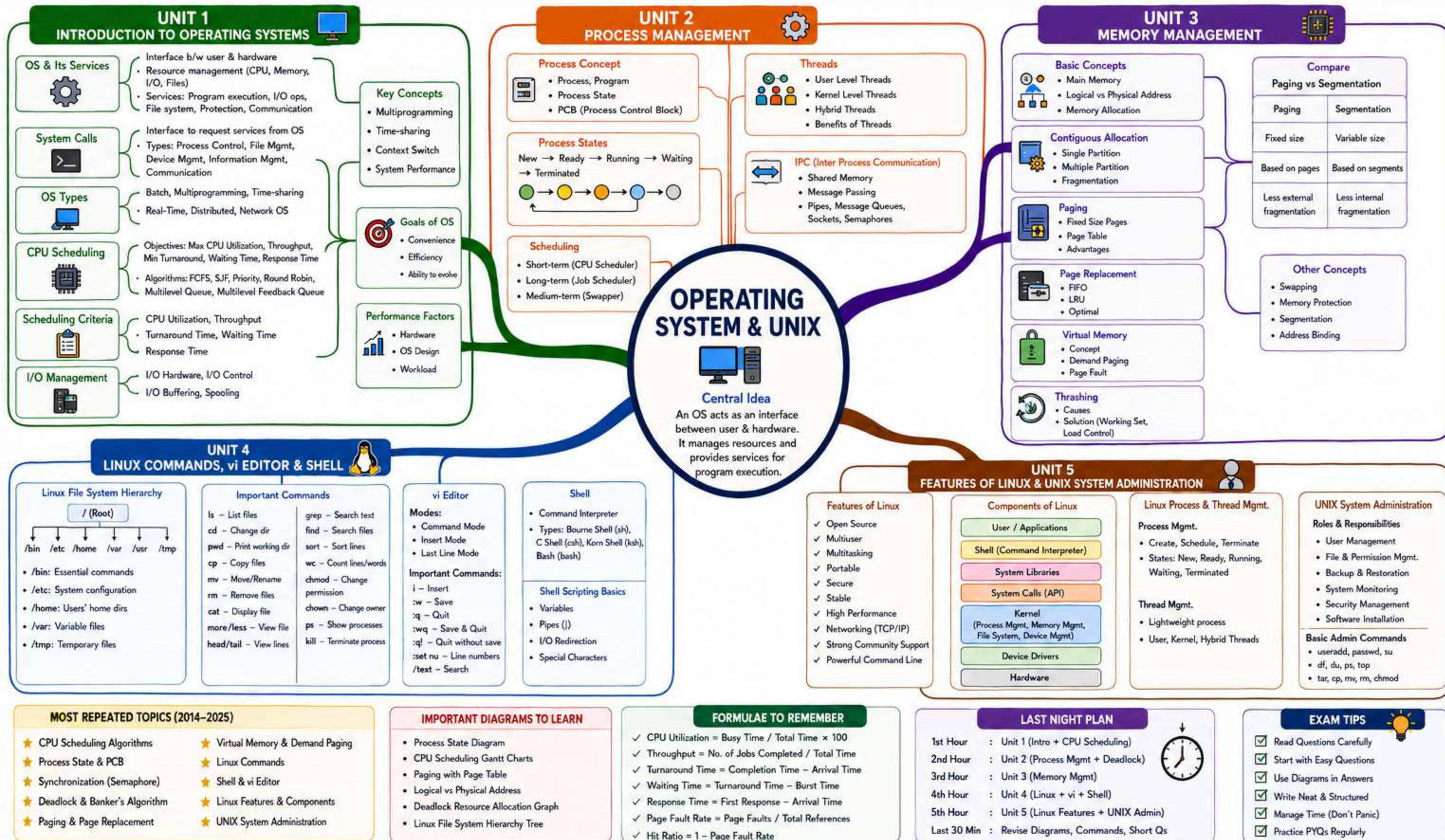
BCA-205 OPERATING SYSTEM & UNIX

MIND MAP OF ALL 5 UNITS

Remember:

- ✓ Understand Concepts
- ✓ Practice PYQs
- ✓ Revise Smartly

Score High
All The Best!



CONCEPT CLEAR → PRACTICE MORE → SCORE HIGH ★ ALL THE BEST FOR 9 JULY TEST!



9 JULY TEST
BE READY!

BCA-205 OPERATING SYSTEM & UNIX

MCQ + ONE-LINER REVISION SHEET

- ✓ Quick Revision
- ✓ High Score
- ✓ All The Best!



UNIT 1: INTRODUCTION TO OPERATING SYSTEM

- An Operating System is:
A) System Software B) Application Software
C) Compiler D) Assembler
- Which of the following is NOT an OS service?
A) Program Execution B) I/O Operations
C) Dancing D) File System Management
- System Calls are used to:
A) Call a system function from user program
B) Increase CPU speed
C) Manage memory D) None of these
- Which scheduling algorithm is non-preemptive?
A) Round Robin B) SJF C) FCFS D) Priority
- Time-sharing system provides each user a small
A) Space B) Memory C) Time slice D) File
- Multiprogramming aims to:
A) Maximize CPU utilization
B) Maximize memory
C) Maximize disk space
D) None of these
- Which of the following is a real-time OS example?
A) Windows B) UNIX C) VxWorks D) Linux
- I/O Buffering is used to:
A) Increase CPU speed B) Increase throughput
C) Reduce response time D) All of these
- Which criteria is used to measure CPU scheduling performance?
A) Turnaround Time B) Waiting Time
C) Response Time D) All of these
- Batch system is an example of:
A) Interactive system B) Real-time system
C) Time-sharing system D) None of these

UNIT 2: PROCESS MANAGEMENT

- A process is:
A) Program in execution B) A file
C) A command D) Data in memory
- Process Control Block (PCB) contains:
A) Process ID only B) Process info
C) Program code D) User data only
- Which state comes after Ready state?
A) Running B) Blocked C) Terminated D) New
- Which of the following is NOT a thread model?
A) User level B) Kernel level
C) Multi level D) Single level
- Critical Section is a part of program where:
A) Shared data is accessed
B) I/O operations are performed
C) New process is created
D) Memory is allocated
- Semaphore is a:
A) Hardware device B) Variable
C) File D) Process
- Deadlock occurs when:
A) Two or more processes wait forever
B) A process terminates
C) Memory is full D) CPU is idle
- Which is NOT a necessary condition for deadlock?
A) Mutual Exclusion B) Hold and Wait
C) No Preemption D) Circular Wait
- Banker's Algorithm is used for:
A) CPU scheduling B) Deadlock detection
C) Deadlock avoidance D) Memory allocation
- Producer-Consumer problem is solved using:
A) Semaphore B) Mutex C) Stack D) Queue

UNIT 3: MEMORY MANAGEMENT

- Memory Management is required for:
A) Efficient use of memory
B) Increase CPU speed
C) Backup purpose
D) None of these
- Paging is a technique of:
A) Memory allocation
B) Secondary storage allocation
C) CPU scheduling
D) None of these
- Page size is:
A) Variable B) Fixed
C) Depends on segments D) None
- Which of the following is a page replacement algorithm?
A) FCFS B) SJF C) FIFO D) RR
- Belady's Anomaly occurs in:
A) FIFO B) LRU C) Optimal D) None
- Virtual memory allows execution of process whose size is:
A) Equal to RAM B) Greater than RAM
C) Less than RAM D) Equal to Disk
- Demand Paging is a type of:
A) Virtual memory B) Cache memory
C) Main memory D) Register memory
- Thrashing occurs when:
A) Too many page faults occur
B) CPU is idle
C) Memory is full
D) Disk is full
- Page Fault occurs when:
A) Page is in memory
B) Page is not in memory
C) CPU is overloaded
D) Disk is full
- Segmentation is based on:
A) Fixed size blocks
B) Logical divisions
C) Physical divisions
D) Virtual addresses

ONE-LINER REVISION (TOP 50)

- OS is system software.
- OS acts as an interface between user and hardware.
- OS provides services to users and programs.
- System call is a request to OS.
- CPU scheduling allocates CPU to processes.
- FCFS is non-preemptive.
- SJF selects process with shortest burst time.
- Round Robin is time-sharing algorithm.
- Priority Scheduling selects by priority.
- Multiprogramming increases CPU utilization.
- Time-sharing gives each user time slice.
- Real-time system responds in real time.
- Throughput is number of jobs completed.
- Turnaround time = Completion - Arrival.
- Waiting time = Turnaround - Burst.
- Process is program in execution.
- PCB stores process information.
- New → Ready → Running → Waiting → Terminated.
- Threads share code, data and resources.
- Critical section accesses shared data.
- Semaphore is an integer variable.
- Mutex allows only one process.
- Deadlock causes processes to wait forever.
- Deadlock needs 4 conditions.
- Banker's Algorithm avoids deadlock.
- Paging divides memory into pages.
- Page size is fixed.
- Page table maps page to frame.
- FIFO may cause Belady's Anomaly.
- LRU replaces least recently used page.
- Virtual memory uses disk as extension.
- Demand paging loads page on demand.
- Thrashing occurs due to too many faults.
- Swapping moves process to disk.
- Segmentation divides by logical units.
- ls lists files and directories.
- cd changes directory.
- cp copies files.
- rm removes files.
- cat displays file content.
- grep searches text in files.
- vi is a screen editor.
- vi has 3 modes: Command, Insert, Last line.
- Shell is command interpreter.
- Bash is default shell in Linux.
- Linux is open source.
- Kernel is core of Linux.
- / is root directory.
- chmod changes permissions.
- chown changes owner of file.

UNIT 4: LINUX COMMANDS, vi EDITOR & SHELL

- Which command is used to list files and directories?
A) list B) dir C) ls D) show
- Command to change directory is:
A) cd B) ch C) dir D) cwd
- Which command is used to copy files?
A) cp B) copy C) mv D) xcopy
- Which command is used to remove files?
A) rm B) del C) remove D) erase
- Which command is used to display file content?
A) type B) cat C) more D) view
- Which command is used to search text in file?
A) find B) grep C) locate D) search
- Which command is used to show current directory?
A) pwd B) cd C) dir D) ls
- vi editor is a:
A) Line editor B) Stream editor
C) Screen editor D) None of these
- In vi editor, to save and quit press:
A) :wq B) :q! C) :x D) :w
- Which shell is the default shell in Linux?
A) Bourne Shell B) C Shell
C) Bash Shell D) Korn Shell

vi Editor Modes (MCQ)

- vi has how many modes?
A) 1 B) 2 C) 3 D) 4
- In which mode we insert text?
A) Command mode
B) Insert mode
C) Last line mode
D) None
- To delete a line in vi we use:
A) dd B) del C) dl D) x

Shell Types (MCQ)

- Which of the following is NOT a shell?
A) Bourne Shell
B) C Shell
C) Bash Shell
D) Python Shell
- Shell is a:
A) Command interpreter
B) Compiler
C) Assembler
D) None

UNIT 5: FEATURES OF LINUX & UNIX SYSTEM ADMINISTRATION

- Linux is:
A) Open Source B) Proprietary
C) Paid Software D) None
- Which component is the heart of Linux system?
A) Shell B) Kernel C) Compiler D) Editor
- Which command is used to create a directory?
A) mkdir B) create C) dir D) make
- Which command is used to change file permission?
A) chmod B) chown C) chgrp D) change
- Which command is used to change file owner?
A) chown B) chmod C) usermod D) own
- The /etc directory contains:
A) Binary files B) Configuration files
C) User files D) Temporary files
- Which command is used to find a file?
A) find B) search C) locate D) where
- The process of taking backup is called:
A) Recovery B) Restoration
C) Backup D) Saving
- Which command shows current date and time?
A) time B) date C) cal D) clock
- Which command is used to view running processes?
A) ps B) top C) jobs D) kill

IMPORTANT FORMULAE

Turnaround Time = Completion Time - Arrival Time
Waiting Time = Turnaround Time - Burst Time
Response Time = First Response - Arrival Time
CPU Utilization = (Busy Time / Total Time) × 100
Throughput = No. of jobs completed / Total Time
hit ratio = 1 - Page Fault Rate

IMPORTANT LINUX COMMANDS (QUICK LIST)

Command	Use	Command	Use
ls	List files & dirs	grep	Search text
cd	Change directory	find	Find files
pwd	Print working directory	mkdir	Create directory
cp	Copy files	rmdir	Remove directory
mv	Move/Rename	chmod	Change permission
rm	Remove files	chown	Change owner
cat	Display file	ps	Show processes
more	View file page by page	top	Process monitor
less	View file (scroll)	df	Disk free space

vi EDITOR COMMANDS

Command	Command
i	Insert before cursor
I	Insert at beginning
a	Insert after cursor
A	Insert at end
O	Open new line below
O	Open new line above
dd	Delete current line
yy	Yank (copy) line
p	Paste after cursor
:w	Save file
:q	Quit
:wq	Save and Quit
:q!	Quit without save
:set nu	Show line numbers
/text	Search forward
?text	Search backward
n	Next match
N	Previous match

FILE PERMISSIONS (chmod)

r = 4 (Read)
w = 2 (Write)
x = 1 (Execute)
Example: chmod 755 file
• 7 = rwx (owner)
5 = r-x (group)
5 = r-x (others)
• Total = 7+5+5 = 17

EXAM TIPS

- ✓ Read questions carefully.
- ✓ Focus on high weightage topics.
- ✓ Practice PYQs.
- ✓ Manage time in exam.
- ✓ Stay calm and confident.

MOST REPEATED TOPICS

- ★ CPU Scheduling Algorithms
- ★ Process State & PCB
- ★ Deadlock & Banker's Algorithm
- ★ Paging & Page Replacement
- ★ Virtual Memory & Demand Paging
- ★ Linux Commands
- ★ vi Editor & Shell
- ★ Linux Features
- ★ UNIX System Administration
- ★ File Permissions

SHORT TRICKS

- ✓ Remember formulas
- ✓ Use diagrams
- ✓ Write step by step
- ✓ Use examples
- ✓ Revise one-liners



ALL THE BEST
FOR 9 JULY TEST!



SCORE HIGH



BCA-205 OPERATING SYSTEM & UNIX

MCQ + ONE-LINER REVISION SHEET (With Answers)

9 JULY TEST
Be Ready!

★ 50 MCQs

🎯 One-Liners

🎯 Exam Focused

🏆 Score High!

UNIT 1: INTRODUCTION TO OS

- OS is system software.
A) True B) False
Ans: A
- Which is not an OS service?
A) Program execution B) Dancing
C) I/O operation D) File management
Ans: B
- System calls are used to:
A) Request OS services B) Manage hardware
C) Increase speed D) None of these
Ans: A
- Which scheduling is non-preemptive?
A) Round Robin B) SJF C) FCFS D) Priority
Ans: C (FCFS)
- Multiprogramming aims to:
A) Maximize CPU utilization
B) Maximize memory C) Increase speed D) None
Ans: A
- Real-time system responds in:
A) Real time B) Batch time
C) User time D) None
Ans: A
- I/O buffering is used to:
A) Increase CPU speed
B) Increase throughput
C) Reduce response time
D) All of these
Ans: D
- Batch system is an example of:
A) Interactive system B) Real-time system
C) Time-sharing system D) None
Ans: C
- CPU scheduling objectives include:
A) Throughput B) Turnaround time
C) Waiting time D) All of these
Ans: D
- Time-sharing system provides each user a small:
A) Space B) Memory
C) Time slice D) File
Ans: C (Time slice)

UNIT 2: PROCESS MANAGEMENT

- A process is:
A) Program in execution B) A file
C) A command D) Data in memory
Ans: A
- PCB contains:
A) Process ID only B) Process info
C) Program code D) User data only
Ans: B
- New state comes after Ready state?
A) Running B) Blocked
C) Terminated D) New
Ans: A (Running)
- Which is NOT a thread model?
A) User level B) Kernel level
C) Multi level D) Single level
Ans: C (Multi level)
- Critical section is a part of program where:
A) Shared data is accessed
B) I/O operations are performed
C) New process is created
D) Memory is allocated
Ans: A
- Semaphore is a:
A) Hardware device B) Variable
C) File D) Process
Ans: B
- Deadlock occurs when:
A) Two or more processes wait forever
B) A process terminates
C) Memory is full D) CPU is idle
Ans: A
- Banker's Algorithm is used for:
A) CPU scheduling B) Deadlock detection
C) Deadlock avoidance D) Memory allocation
Ans: C
- Producer-Consumer problem is solved using:
A) Semaphore B) Mutex C) Stack D) Queue
Ans: A (Semaphore)
- Context switch is:
A) PCB change B) Memory allocation
C) I/O operation D) Device switch
Ans: A

UNIT 3: MEMORY MANAGEMENT

- Memory management is required for:
A) Efficient memory use
B) Increase CPU speed
C) Backup purpose D) None of these
Ans: A
- Paging is a technique of:
A) Memory allocation
B) Secondary storage allocation
C) CPU scheduling D) None
Ans: A
- Page size is:
A) Variable B) Fixed
C) Depends on segments D) None
Ans: B (Fixed)
- Page replacement algorithm:
A) FCFS B) SJF C) FIFO D) RR
Ans: C (FIFO)
- Belady's Anomaly occurs in:
A) FIFO B) LRU C) Optimal D) None
Ans: A (FIFO)
- Virtual memory allows execution of process whose size is:
A) Equal to RAM B) Greater than RAM
C) Less than RAM D) Equal to Disk
Ans: B
- Demand paging is a type of:
A) Virtual memory B) Cache memory
C) Main memory D) Register memory
Ans: A
- Thrashing occurs when:
A) Too many page faults occur
B) CPU is idle C) Memory is full
D) Disk is full
Ans: A
- Page Fault occurs when:
A) Page is in memory
B) Page is not in memory
C) CPU is overloaded D) Disk is full
Ans: B
- Segmentation is based on:
A) Fixed size blocks B) Logical divisions
C) Physical divisions D) Virtual addresses

UNIT 4: LINUX COMMANDS, vi EDITOR & SHELL

- Which command is used to list files?
A) ls B) dir C) cd D) show
Ans: A (ls)
- Command to change directory:
A) cd B) ch C) dir D) cwd
Ans: A (cd)
- Which command copies files?
A) cp B) copy C) mv D) xcopy
Ans: A (cp)
- Command to remove files?
A) rm B) del C) remove D) erase
Ans: A (rm)
- Which command displays file content?
A) type B) cat C) more D) view
Ans: B (cat)
- Command to search text in file?
A) find B) grep C) locate D) search
Ans: B (grep)
- Command used to show current directory?
A) pwd B) cd C) dir D) ls
Ans: A (pwd)
- vi editor is a:
A) Line editor B) Stream editor
C) Screen editor D) None
Ans: A (Line editor)
- Default shell in Linux?
A) Bourne Shell B) C Shell
C) Bash Shell D) Korn Shell
Ans: A (Bourne Shell)
- Which is a Shell Script file extension?
A) .txt B) .sh C) .py D) .exe
Ans: B (.sh)

vi editor Modes

- vi has how many modes?
A) 1 B) 2 C) 3 D) 4
Ans: C (3)
- In which mode we insert text?
A) Command mode
B) Insert mode
C) Last line mode D) None
Ans: B (Insert mode)
- To delete a line in vi we use:
A) dd B) del C) dl D) x
Ans: A (dd)
- To save and quit press:
A) :wq B) :q! C) :x D) :w
Ans: A (:wq)
- Default shell prompt ends with:
A) # B) \$ C) % D) @
Ans: B (\$)

UNIT 5: FEATURES OF LINUX & UNIX SYSTEM ADMINISTRATION

- Linux is:
A) Open Source B) Proprietary
C) Paid Software D) None
Ans: A
- Which is the heart of Linux system?
A) Shell B) Kernel C) Compiler D) Editor
Ans: B (Kernel)
- Command to create a directory?
A) mkdir B) create C) dir D) make
Ans: A (mkdir)
- Command to change file permission?
A) chmod B) chown C) chgrp D) change
Ans: A (chmod)
- Which command shows current date & time?
A) time B) date C) cal D) clock
Ans: B (date)

IMPORTANT LINUX COMMANDS (Quick List)

Command	Use
ls	List files & directories
cd	Change directory
pwd	Print working directory
cp	Copy files
mv	Move/Rename files
rm	Remove files
cat	Display file content
more	View file page by page
grep	Search text in files
chmod	Change permissions
chown	Change owner
ps	Show processes
kill	Terminate process

vi EDITOR COMMANDS

Command	Use
i	Insert before cursor
a	Insert after cursor
o	Open new line
dd	Delete current line
yy	Copy (yank) line
p	Paste after cursor
:w	Save file
:q	Quit
:wq	Save & Quit
:q!	Quit without save
:set nu	Show line numbers

ONE-LINER REVISION (Top 15)

- OS is system software. → True
- OS acts as interface. → True
- OS provides services. → True
- System call is request to OS. → True
- CPU scheduling allocates CPU. → True
- FCFS is non-preemptive. → True
- Round Robin uses time slice. → True
- SJF selects shortest job. → True
- Multitasking increases CPU use. → True
- Time-sharing gives small time slice. → True
- Virtual memory > physical memory. → True
- Page size is fixed. → True
- Thrashing occurs by many faults. → True
- Deadlock needs 4 conditions. → True
- Linux is open source. → True

FILE PERMISSIONS (chmod)

r = 4 (Read)
w = 2 (Write)
x = 1 (Execute)
Example:
• **chmod 755 file**
7 = rwx (owner)
5 = r-x (group)
5 = r-x (others)

EXAM TIPS

- ✓ CPU Utilization = $\frac{\text{Total Time}}{\text{Total Time} + \text{Idle Time}} \times 100$
- ✓ Focus on High Weightage
- ✓ Practice PYQs
- ✓ Manage Time Wisely
- ✓ Stay Calm & Confident

★ CONCEPT CLEAR → PRACTICE MORE → SCORE HIGH

All The Best for 9 July Test!



HIGH WEIGHTAGE TOPICS

- ✓ CPU Scheduling
- ✓ Process & Threads
- ✓ Paging & Memory
- ✓ Linux Commands
- ✓ UNIX Administration

UNIT 1: INTRODUCTION TO OS	UNIT 2: PROCESS MANAGEMENT	UNIT 3: MEMORY MANAGEMENT	ONE-LINER REVISION (TOP 50) WITH ANSWERS
1. An Operating System is: Ans: A) System Software 2. Which of the following is NOT an OS service? Ans: C) Dancing 3. System Calls are used to: Ans: A) Call a system function from user program 4. Which scheduling algorithm is non-preemptive? Ans: A) Round Robin 5. Time-sharing system provides each user a small: Ans: C) Time slice 6. Multiprogramming aims to: Ans: A) Maximize CPU utilization 7. Which of the following is a real-time OS example? Ans: B) UNIX 8. I/O Buffering is used to: Ans: A) Increase CPU speed 9. Which criteria is used to measure CPU scheduling performance? Ans: A) Turnaround Time 10. Batch system is an example of: Ans: B) Real-time system	1. A process is: Ans: A) Program in execution 2. PCB contains: Ans: A) Process ID only 3. Which state comes after Ready state? Ans: A) Running 4. Which of the following is NOT a thread model? Ans: C) Multi level 5. Critical Section is a part of program where: Ans: B) I/O operations are performed 6. Semaphore is a: Ans: A) Hardware device 7. Deadlock occurs when: Ans: B) A process terminates 8. Which is NOT a necessary condition for deadlock? Ans: B) Hold and Wait 9. Banker's Algorithm is used for: Ans: B) Deadlock detection 10. Producer-Consumer problem is solved using: Ans: A) Semaphore	1. Memory Management is required for: Ans: A) Efficient use of memory 2. Paging is a technique of: Ans: A) Memory allocation 3. Page size is: Ans: A) Variable 4. Which of the following is a page replacement algorithm? Ans: A) FCFS 5. Belady's Anomaly occurs in: Ans: B) LRU 6. Virtual memory allows execution of process whose size is: Ans: A) Equal to RAM 7. Demand Paging is a type of: Ans: A) Virtual memory 8. Thrashing occurs when: Ans: A) Too many page faults occur 9. Page Fault occurs when: Ans: B) Page is not in memory 10. Segmentation is based on: Ans: C) Physical divisions	1. OS is system software. Ans: True 2. OS acts as an interface. Ans: True 3. OS provides services to users. Ans: True 4. System call is a request to OS. Ans: True 5. CPU scheduling allocates CPU. Ans: True 6. FCFS is non-preemptive. Ans: True 7. SJF selects shortest job. Ans: True 8. Round Robin is time-sharing. Ans: True 9. Priority scheduling selects by priority. Ans: True 10. Multiprogramming increases CPU use. Ans: True 11. Time-sharing gives each user time slice. Ans: True 12. Real-time system responds in real time. Ans: True 13. Throughput is number of jobs done. Ans: True 14. Turnaround time = Completion - Arrival. Ans: True 15. Waiting time = Turnaround - Burst. Ans: True 16. Process is program in execution. Ans: True 17. PCB stores process information. Ans: True 18. New → Ready → Running → Waiting → Terminated. Ans: True 19. Threads share code, data & resources. Ans: True 20. Critical section accesses shared data. Ans: True 21. Semaphore is an integer variable. Ans: True 22. Mutex allows only one process. Ans: True 23. Deadlock causes processes to wait forever. Ans: True 24. Deadlock needs 4 conditions. Ans: True 25. Banker's Algorithm avoids deadlock. Ans: True 26. Paging divides memory into pages. Ans: True 27. Page size is fixed. Ans: True 28. Page table maps page to frame. Ans: True 29. FIFO may cause Belady's Anomaly. Ans: True 30. LRU replaces least recently used page. Ans: True 31. Virtual memory uses disk as extension. Ans: True 32. Demand paging loads page on demand. Ans: True 33. Thrashing occurs due to too many faults. Ans: True 34. Swapping moves process to disk. Ans: True 35. Segmentation divides by logical units. Ans: True 36. ls lists files and directories. Ans: True 37. cd changes directory. Ans: True 38. cp copies files. Ans: True 39. rm removes files. Ans: True 40. cat displays file content. Ans: True 41. grep searches text in files. Ans: True 42. vi is a screen editor. Ans: True 43. vi has 3 modes: Command, Insert, Last line. Ans: True 44. Shell is command interpreter. Ans: True 45. Bash is default shell in Linux. Ans: True 46. Linux is open source. Ans: True 47. Kernel is core of Linux. Ans: True 48. / is root directory. Ans: True 49. chmod changes permissions. Ans: True 50. chown changes owner of file. Ans: True

UNIT 4: LINUX COMMANDS, vi EDITOR & SHELL	UNIT 5: FEATURES OF LINUX & UNIX SYSTEM ADMINISTRATION
LINUX COMMANDS (MCQ) 1. Which command is used to list files & dirs? Ans: A) ls 2. Command to change directory is: Ans: C) cd 3. Which command is used to copy files? Ans: B) cp 4. Which command is used to remove files? Ans: C) rm 5. Which command is used to display file content? Ans: A) cat 6. Which command is used to search text in file? Ans: B) grep 7. Which command is used to show current directory? Ans: C) pwd 8. vi editor is a: Ans: A) Line editor 9. In vi editor, to save and quit press: Ans: B) :q! 10. Default shell in Linux is: Ans: B) Bash Shell vi EDITOR MODES (MCQ) 1. vi has how many modes? Ans: B) 3 2. In which mode we insert text? Ans: A) Command mode 3. To delete a line in vi we use: Ans: A) dd SHELL TYPES (MCQ) 1. Which of the following is NOT a shell? Ans: D) Python Shell 2. Shell is a: Ans: A) Command interpreter 3. Which shell is Bourne Shell? Ans: A) Bourne Shell	1. Linux is: Ans: A) Open Source 2. Which component is the heart of Linux system? Ans: C) Kernel 3. Which command is used to create a directory? Ans: A) mkdir 4. Which command is used to change file permission? Ans: D) chmod 5. Which command is used to change file owner? Ans: B) chown

UNIX SYSTEM ADMINISTRATION (MCQ)

6. The /etc directory contains:
Ans: B) Configuration files

7. Which command is used to find a file?
Ans: A) find

8. The process of taking backup is called:
Ans: B) Restoration

9. Which command shows current date and time?
Ans: B) date

IMPORTANT FORMULAE	IMPORTANT LINUX COMMANDS (QUICK LIST)				vi EDITOR COMMANDS				FILE PERMISSIONS (chmod)	
• Turnaround Time = Completion Time – Arrival Time • Waiting Time = Turnaround Time – Burst Time • Response Time = First Response – Arrival Time • CPU Utilization = (Busy Time / Total Time) × 100 • Throughput = No. of jobs completed / Total Time • hit ratio = 1 – Page Fault Rate	Command	Use	Command	Use	Command	Use		Use		
	ls	List files & dirs	grep	Search text	i	Insert before cursor	:w	Save file	r	= 4 (Read)
	cd	Change directory	find	Find files	I	Insert at beginning	:q	Quit	w	= 2 (Write)
	pwd	Print working directory	mkdir	Create directory	a	Insert after cursor	:wq	Save and Quit	x	= 1 (Execute)
	cp	Copy files	rmdir	Remove directory	A	Insert at end	:q!	Quit without save	Example: chmod 755 file	
	mv	Move /Rename	chmod	Change permission	o	Open new line below	:set nu	Show line numbers	• 7 = rwx (owner)	
	rm	Remove files	chown	Change owner	O	Open new line above	/text	Search forward	5 = r-x (group)	
	cat	Display file	ps	Show processes	dd	Delete current line	?text	Search backward	5 = r-x (others)	
	more	View file page by page	top	Process monitor	yy	Yank (copy) line	n	Next match	• Total = 7+5+5 = 17	
	less	View file (scroll)	df	Disk free space	p	Paste after cursor	N	Previous match		

EXAM TIPS	MOST REPEATED TOPICS (2014-2025)	SHORT TRICKS	ALL THE BEST FOR 9 JULY TEST! SCORE HIGH
✓ Read questions carefully. ✓ Focus on high weightage topics. ✓ Practice PYQs. ✓ Manage time in exam. ✓ Stay calm and confident.	★ CPU Scheduling Algorithms ★ Process State & PCB ★ Deadlock & Banker's Algorithm ★ Paging & Page Replacement ★ Virtual Memory & Demand Paging ★ Linux Commands ★ vi Editor & Shell ★ Linux Features ★ UNIX System Administration ★ File Permissions	✓ Remember formulas ✓ Use diagrams ✓ Write step by step ✓ Use examples ✓ Revise one-liners	★ ★ ★ ★

UNIT 1: INTRODUCTION TO OS

1. Q. Explain Operating System and its services.
Ans: An OS is system software that acts as an interface between user and hardware. Services:
1. Program execution 2. I/O operations
3. File system management 4. Memory management
5. Error detection 6. Resource allocation
7. Protection 8. Accounting

2. Q. What are System Calls? Explain types.
Ans: System calls are requests made by user programs to OS kernel.
Types: 1. Process control 2. File management
3. Device management 4. Information maintenance 5. Communication

3. Q. Explain CPU Scheduling and its objectives.
Ans: CPU scheduling selects a process from ready queue to execute. Objectives:
1. High CPU utilization 2. High throughput
3. Low turnaround time 4. Low waiting time
5. Low response time

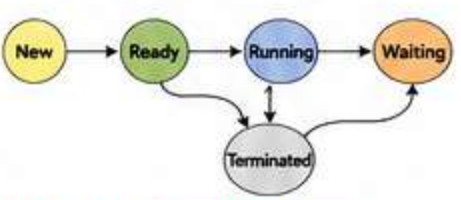
4. Q. What is Multiprogramming?
Ans: When multiple programs are loaded in memory and CPU switches among them to maximize CPU utilization.

5. Q. What is Time-sharing system?
Ans: CPU time is divided into small time slices and each process gets a turn. It provides interactive computing.

6. Q. Explain Scheduling Criteria.
Ans: 1. CPU Utilization – keep CPU busy
2. Throughput – max. no. of jobs/time
3. Turnaround Time – completion – arrival
4. Waiting Time – total time in ready queue
5. Response Time – first response – arrival

UNIT 2: PROCESS MANAGEMENT

1. Q. Explain Process and Process State Diagram.
Ans: A process is a program in execution.
States: New → Ready → Running → Waiting → Terminated (with possible transitions).



2. Q. What is a Thread? Explain types.
Ans: A thread is smallest unit of execution.
Types: 1. User Level Threads
2. Kernel Level Threads
3. Hybrid Threads

3. Q. What is Critical Section Problem?
Ans: When multiple processes access shared data and result depends on timing, it is called Critical Section Problem.

4. Q. Explain Semaphore.
Ans: Semaphore is an integer variable used for process synchronization.
Types: 1. Binary Semaphore (value 0 or 1)
2. Counting Semaphore (0 to n)

5. Q. What is Deadlock? Explain conditions.
Ans: Deadlock is a situation where processes are waiting for resources held by each other.
Conditions (Coffman Conditions):
1. Mutual Exclusion 2. Hold and Wait
3. No Preemption 4. Circular Wait

6. Q. Explain Banker's Algorithm.
Ans: Banker's Algorithm is used to avoid deadlock. It checks if system is in safe state by using Available, Max, Allocation, Need matrices.

UNIT 3: MEMORY MANAGEMENT

1. Q. Explain Paging with neat diagram.
Ans: Paging is a memory management scheme in which logical memory is divided into pages and physical memory into frames. Page table maps page no. to frame no.

Logical Memory	Page Table	Physical Memory
Page 0	Page no. 0	Frame 0
Page 1	Frame no. 3	Frame 1
.	1	6
.	.	.
Page n	n	f
		Frame f

2. Q. Explain Page Replacement Algorithms.
Ans:
• FIFO – Replace oldest page.
• LRU – Replace least recently used page.
• Optimal – Replace page not used for longest future time.

3. Q. What is Paging vs Segmentation?
Ans:

Paging	Segmentation
Fixed size	Variable size
Based on pages	Based on segments
Less external fragmentation	Less internal fragmentation

4. Q. What is Virtual Memory?
Ans: It is a memory management scheme that uses secondary memory to extend RAM.

5. Q. What is Thrashing? How to control?
Ans: Thrashing occurs when system spends more time in paging than execution.
Control: 1. Increase RAM
2. Working Set Model 3. Page Fault Frequency

UNIT 4: LINUX COMMANDS, SHELL & vi EDITOR

1. Q. What is Shell? Explain types.
Ans: Shell is command interpreter.
Types: 1. Bourne Shell (sh)
2. C Shell (csh)
3. Korn Shell (ksh)
4. Bash Shell (bash)

2. Q. Write syntax of I/O Redirection.
Ans:
• Output Redirection: command > file
• Input Redirection: command < file
• Append: command >> file
• Error Redirection: command 2> file
• Both: command > file 2>&1

3. Q. What is Pipe? Give example.
Ans: Pipe (|) is used to pass output of one command as input to another.
Example: ls -l | more

4. Q. What is vi editor? Explain modes.
Ans: vi is a text editor.
Modes: 1. Command mode (Esc)
2. Insert mode (i)
3. Last line mode (:)

5. Q. Write vi commands.
Ans:
• i – Insert • dd – Delete line
• :w – Save • :q – Quit
• :wq – Save & Quit • :q! – Quit without save
• /text – Search • n – Next match

6. Q. Write some important Linux commands.
Ans:
• ls – list files • cp – copy files
• mv – move files • rm – remove files
• cat – display file • more – view file
• grep – search text • chmod – change permission
• chown – change owner • pwd – print working dir
• df – disk free • du – disk usage

UNIT 5: FEATURES OF LINUX & UNIX SYSTEM ADMINISTRATION

1. Q. What are features of Linux?
Ans: 1. Open Source 2. Multiuser
3. Multitasking 4. Portable
5. Secure 6. Stable 7. High Performance
8. Networking 9. Strong Community Support
10. Powerful Command Line

2. Q. What are components of Linux?
Ans: 1. Kernel 2. Shell 3. System Libraries
4. System Calls 5. Utilities 6. File System
7. Device Drivers 8. Applications

3. Q. Explain Linux Process Management.
Ans: Linux uses fork() to create process and exec() to replace process image. Commands: ps, top, kill, nice, renice.

4. Q. Explain Linux Thread Management.
Ans: Linux supports both User Level Threads (ULT) and Kernel Level Threads (KLT).
Commands: ps -L, top -H.

5. Q. What is the role of UNIX System Administrator?
Ans: 1. User management 2. File & permission
3. Backup & restore 4. System monitoring
5. Security management 6. Software installation
7. Performance tuning

6. Q. Explain Backup and Restoration in UNIX.
Ans: Backup is copying data to prevent loss.
Commands: tar, cp, dump. Restore is recovering data from backup.
Commands: tar -xvf, restore, cp.

TOP EXPECTED PYQS (2014-2025) SOLVED			
No.	Question (Year)	Answer (Short)	Marks
1	Explain CPU Scheduling algorithms. (2014, 2016, 2018, 2021)	FCFS, SJF, Priority, RR with example	8
2	Explain Process State Diagram. (2015, 2017, 2019, 2022)	New, Ready, Running, Waiting, Terminated with transitions	5
3	Explain Deadlock and conditions. (2014, 2016, 2018, 2020, 2023)	Deadlock is waiting for resources. Conditions: M.E., H&W, N.P., C.W.	5
4	Explain Banker's Algorithm. (2015, 2017, 2019, 2024)	Uses Available, Max, Allocation, Need matrices to check safe state.	8
5	Explain Paging with diagram. (2014, 2015, 2017, 2019, 2021)	Logical memory → pages, Physical memory → frames, page table mapping.	5
6	Explain Page Replacement algorithms. (2016, 2018, 2020, 2022, 2024)	FIFO, LRU, Optimal with example.	8
7	What is Virtual Memory? (2015, 2018, 2021, 2023)	Uses secondary memory to extend primary memory.	5
8	What is Thrashing? How to control? (2016, 2019, 2022, 2024)	Occurs when paging > execution. Control: More RAM, Working Set.	5
9	Explain Shell and types. (2014, 2016, 2018, 2021)	Shell is command interpreter. Types: sh, csh, ksh, bash.	5
10	Explain I/O Redirection with examples. (2015, 2017, 2020, 2023)	>, <, >>, 2>, 2>&1 with examples.	5
11	Explain Pipe with example. (2016, 2018, 2021, 2024)	Pipe () passes output of one command to another.	3
12	Explain vi editor modes and commands. (2014, 2015, 2017, 2019, 2022)	Modes: Command, Insert, Last line. Commands: i, dd, :w, :q, :wq, :q!	5
13	Explain features of Linux. (2014, 2016, 2019, 2023)	Open Source, Multiuser, Portable, Secure, Stable, etc.	5
14	Explain components of Linux. (2015, 2018, 2020, 2024)	Kernel, Shell, Libraries, System Calls, Utilities, File System, Drivers, Apps.	5
15	Explain role of UNIX System Administrator. (2016, 2017, 2019, 2022, 2025)	User mgmt, File & permission, Backup, Monitor, Security, Install, Tune.	5

ONE-LINER REVISION (TOP 25)

1. OS is system software.

2. OS acts as an interface.

3. System call is request to OS.

4. CPU scheduling allocates CPU.

5. FCFS is non-preemptive.

6. Round Robin uses time slice.

7. SJF selects shortest job.

8. Process is a program in execution.

9. Thread is smallest unit of execution.

10. Semaphore is integer variable.

11. Deadlock needs 4 conditions.

12. Paging uses page table.

13. Fragmentation in paging is less.

14. Segmentation uses variable size.

15. Virtual memory extends RAM.

16. Thrashing occurs in paging.

17. Shell is command interpreter.

18. Pipe uses | symbol.

19. vi has 3 modes.

20. In vi, i is insert mode.

21. :wq saves and quits.

22. ls lists files.

23. grep searches text.

24. chmod changes permission.

25. Linux is open source.

LINUX COMMANDS QUICK REFERENCE			
Command	Use	Command	Use
ls	List files	chmod	Change permissions
pwd	Print working directory	chown	Change owner
cd	Change directory	df	Disk free space
cp	Copy files	du	Disk usage
mv	Move/Rename files	ps	Process status
rm	Remove files	kill	Terminate process
cat	Display file content	top	Active processes
more	View file (page by page)	tar	Backup/Archive
less	View file (scroll)	man	Manual pages
grep	Search text in files	clear	Clear screen

IMPORTANT FORMULAE	
• CPU Utilization	= Busy Time / Total Time × 100
• Throughput	= No. of jobs completed / Total Time
• Turnaround Time	= Completion Time – Arrival Time
• Waiting Time	= Turnaround Time – Burst Time
• Response Time	= First Response – Arrival Time
• Page Fault Rate	= Page Faults / Total References
• Hit Ratio	= 1 – Page Fault Rate

EXAM TIPS	
✓	Focus on CPU Scheduling, Deadlock, Paging.
✓	Learn Linux commands by practice.
✓	Revise vi editor commands.
✓	Solve old papers for better score.
✓	Manage time and write neat answers.

MARKS DISTRIBUTION	
Unit 1	: 10-15
Unit 2	: 15-20
Unit 3	: 15-20
Unit 4	: 15-20
Unit 5	: 10-15
Total	: 75-80 Marks